NONLINEAR DIFFERENCE-IN-DIFFERENCES FOR MANIFOLD DATA VIA OPTIMAL TRANSPORT

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Abstract. This paper studies difference-in-differences (DID) for outcomes that take values in a Riemannian manifold, such as vote-share compositions. Existing approaches, including Geodesic DID, restrict an average movement of the outcome and thus require the average movements of the treated and control groups to agree in both size and direction. When the untreated movement depends on the baseline position and the two groups start from different positions, the implied counterfactual can therefore point in a counterintuitive direction. We develop manifold changes-in-changes (MCiC), whose target is the untreated counterfactual distribution of the treated group. MCiC assumes a common untreated transition map, which moves each baseline position on a common tangent-space chart to its untreated post-treatment position in the same way for both groups. We set out conditions under which the optimal transport between the pre-treatment and post-treatment distributions of the control group identifies this map, so that applying it to the treated baseline distribution recovers the counterfactual distribution. We propose a one-nearest-neighbor optimal transport estimator and derive a general-dimensional upper bound on its convergence rate. In simulations, its performance is stable across transition intensities, whereas the errors of mean-based DID grow as the untreated movement toward a central composition strengthens. In an application to early voting in U.S. presidential elections, the estimator gives a negative estimated effect on the share of the residual Other category, whereas mean-based and compositional DID give positive estimated effects.

1. Introduction

Difference-in-differences (DID) is an important method in econometrics and statistics that studies the effect of a treatment by comparing how the outcomes of a treated group and a control group change from before to after the treatment. The treated group receives the treatment in the post-treatment period $t = 1$ but not in the pre-treatment period $t = 0$, and the object of interest is its outcome at $t = 1$ had it not received the treatment. The usual mean-based DID recovers the mean of this outcome through the parallel trends assumption, which states that, without the treatment, the change in the mean outcome of the treated group would have equaled the change in the mean

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outcome of the control group. This assumption allows one to identify and estimate the average treatment effect on the treated. The assumption may be straightforward when the outcome is a scalar, but it takes a nontrivial form when the outcome is not a scalar.

Some outcomes have a geometric structure, and among them this paper considers outcomes that take values in a finite-dimensional Riemannian manifold. Such outcomes include compositional data. In compositional data, for example, each component is nonnegative and the components sum to one, so treating each component as a separate scalar does not capture the structure of the composition as a whole. Adding an ordinary additive mean change to each component can produce a counterfactual composition with a negative component, and the change therefore has to be represented in a way that respects the domain of the outcome and the chosen geometric structure. Difference-in-differences in such settings has been studied by Zhou et al. (2026) and Boussim (2026), among others. In particular, Zhou et al. (2026) develop the Geodesic DID (GDiD), which studies treatment effects for outcomes in geodesic metric spaces and considers parallel trends along the geodesics of the space. On a Riemannian manifold, GDiD parallel transports the movement of the control group's Fréchet mean to the treated group's baseline Fréchet mean and thereby constructs the treated group's untreated post-treatment Fréchet mean. The Fréchet mean of a manifold-valued outcome is a point that minimizes the expected squared geodesic distance to the outcome, and in a Euclidean space it coincides with the ordinary mean. Alternatively, the geometric structure can be preserved by mapping the outcomes into a common tangent-space chart, imposing an additive parallel-trend restriction on the means of the chart coordinates, and mapping the resulting mean back to the manifold. We refer to this construction as Tangent DID (TDiD). Boussim (2026) develops compositional DID (CoDiD), which is designed specifically for compositional data rather than for general manifolds.

All of these approaches restrict an average movement, defined for instance along the geodesics of the manifold or by vector addition in a tangent space, but such a restriction need not produce natural movements on the manifold. Specifically, a parallel movement of this kind has several implications at once, since it requires the average movements of the two groups to agree not only in size but also in direction. A realistic example comes from election data. Consider the vector of party vote shares in each electoral district in a two-party system with Party A and Party B. When vote shares regress to the mean across elections, the direction of the time trend is likely to depend on where each district is at $t = 0$. A district in which Party A has a high share at $t = 0$ would move toward a central composition at $t = 1$, and so would a district in which Party B has a high share

at $t = 0$, as the thin light blue arrows in Figure 1 illustrate. Hence, when the treated and control groups are located at different positions at $t = 0$, the counterfactual implied by parallel trends can point in a counterintuitive direction.

Figure 1 isolates, in the voting design used later in the paper, the additional directional content that the mean parallel-trend restrictions of Geodesic DID and Tangent DID, which amount to transporting the movement of a single mean, can impose on vote-share compositions. In other words, it is an example in which carrying the movement of the control group's mean from $\mu_{C,0}$ at $t = 0$ to $\mu_{C,1}$ at $t = 1$ over to the treated group in parallel yields a counterfactual that does not agree with the untreated movement of the treated group. In this example, the control group starts from a larger Party B share than the central composition toward which the thin light blue arrows point, so its mean moves mainly in the direction in which the Party B share falls. The treated group starts from a larger Party A share than that central composition, so its mean would move mainly in the direction in which the Party A share falls, while the Party B share rises slightly. Because Geodesic DID and Tangent DID carry the movement of the control group over to the baseline mean of the treated group, they imply a counterfactual in which the Party B share of the treated group falls, as panels (a) and (b) of Figure 1 show, and this is opposite to the direction in which that share moves without the treatment.

This concern is not limited to elections and can also arise for other compositional data that are important in economics. One example is the homogenization of regional industrial structure, in which the industry shares of employment in different regions become closer to one another. Kim (1995), for instance, reports that the regional specialization of U.S. manufacturing declined between the 1930s and 1987. Another example is the homogenization of national food supplies, in which the crop compositions of countries, such as the shares of calorie supply by crop, become more similar to one another. Khoury et al. (2014) report such a homogenization of food supplies over 1961–2009. The crop composition of a country can be treated as compositional data in the same way as our vote shares. Other examples of manifold-valued outcomes are periodic quantities, such as the time of day or the time of year, represented on a circle, and several such quantities represented jointly on a torus. For these outcomes as well, the same concern as with the election data can arise, in particular when the outcome moves so as to concentrate somewhere, or more generally when its movement differs by location.

To represent such movements, which depend on the initial position, this paper assumes a common untreated transition map, such as the one drawn by the thin light blue arrows in Figure 1. The

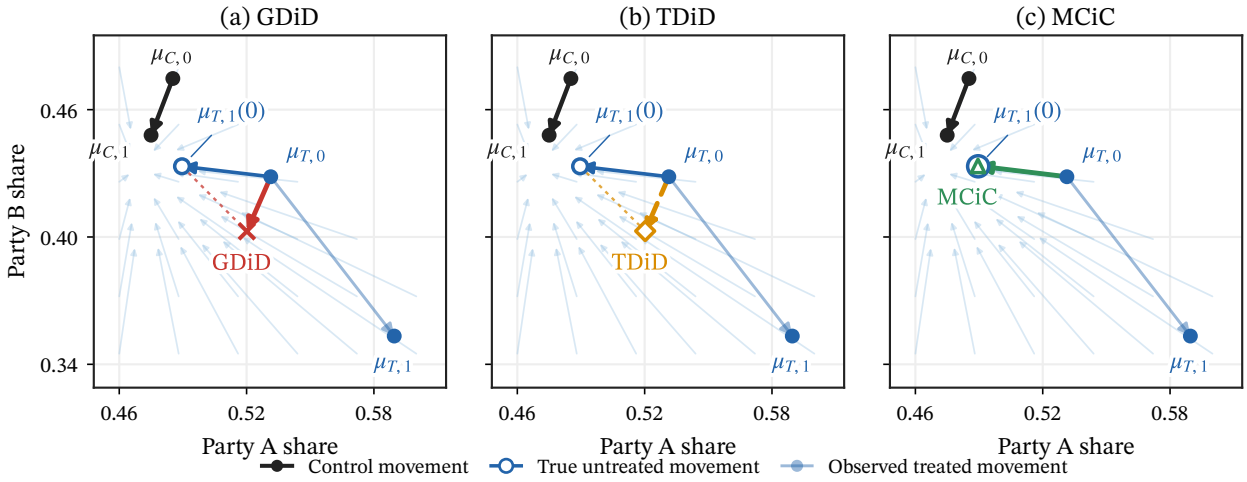


FIGURE 1. Mean movements and counterfactuals in a setting based on the simulation design of Section 6 with transition intensity $\kappa = 0.15$. Each composition in this design consists of the three categories Party A, Party B, and Other, and each panel plots only the shares of Party A and Party B, which reduces the composition to two dimensions. The thin light blue arrows show the untreated movement of the design, which is the same in the treated and control groups and moves each composition toward the composition in which Other has the share 0.10 and Party A has the share 0.52 of the two-party vote, that is, toward the Party A, Party B, and Other shares 0.468, 0.432, and 0.10.

common untreated transition map is the map on a common tangent-space chart that moves a baseline position to the corresponding untreated post-treatment position and acts in the same way on the treated and control groups. That this map is common to the two groups is essential, because it allows us to learn the untreated time evolution, which depends on the baseline position, from the data of the control group and to apply it to the baseline distribution of the treated group to recover the counterfactual distribution.

Building on this assumption, this paper develops manifold changes-in-changes (MCiC), whose target is the untreated counterfactual distribution of the treated group in the post-treatment period. Under the identification conditions of Section 5.1, MCiC identifies the common untreated transition map by the optimal transport between the pre-treatment and post-treatment distributions of the control group. When the counterfactual distribution recovered with the identified map has a unique Fréchet mean, the untreated counterfactual Fréchet mean $\mu_{T,1}(0)$ is obtained as well. In the voting

example, panel (c) of Figure 1 shows that the Fréchet mean of the counterfactual distribution estimated in this way lies close to $\mu_{T,1}(0)$.

This construction, which models the untreated time evolution on a common tangent-space chart by the common untreated transition map, resembles the changes-in-changes model of Athey and Imbens (2006). As in Torous, Gunsilius, and Rigollet (2024), who extend the changes-in-changes model to multivariate outcomes, the transition of the control group identifies the Brenier optimal transport map. The theoretical contribution of this paper is to set out the assumptions under which changes-in-changes can be carried out with manifold-valued data.

The primary estimator, one-nearest-neighbor optimal transport (1NN OT), finds an optimal transport plan between the pre-treatment and post-treatment empirical distributions of the control group and computes the destination of each control baseline observation as the weighted average of the post-treatment points that the plan assigns to it. Each treated baseline observation then receives the destination of its nearest control baseline observation in the common chart, and mapping these destinations back to the manifold gives the estimate of the counterfactual distribution. As in the multivariate Euclidean case, we use the results of Manole et al. (2024) to derive a general-dimensional upper bound on the convergence rate of this estimator under conditions that include independent samples and a chart fixed in advance.

In the simulations, we consider a hypothetical setting with two major parties and a residual category of other candidates. The performance of 1NN OT (Time), the primary estimator of MCiC, is stable over the range of transition intensities that we examine. When untreated vote shares follow the common untreated transition map as in Figure 1, apart from independent shocks in the post-treatment period, the estimation errors of Geodesic DID and Tangent DID grow as the transition becomes stronger.

In the empirical application, we analyze data from the United States, where two major parties dominate. Following the four-state comparison design of Boussim (2026), we examine the effect of early voting. The data are the county-level presidential election returns of the MIT Election Data and Science Lab (MEDSL), distributed through Harvard Dataverse as MIT Election Data and Science Lab (2018). Because we keep the party labels recorded by MEDSL and collapse them into Democrat, Republican, and Other, the results for Other are interpreted as results for the residual category recorded in these data. We compare counties in the treated states, Maryland and New Jersey, with counties in the control states, New York and Pennsylvania. The baseline year is

2004 and the post-treatment year is 2008. The 2000 election serves as an additional pre-treatment observation, and we analyze the 174 counties observed in all of the 2000, 2004, and 2008 elections.

For the pre-treatment period from 2000 to 2004, the mean parallel-trend restrictions of GDiD and TDiD are both rejected at the 5% level. We retain GDiD and TDiD in the comparison as benchmarks whose corresponding pre-treatment mean restrictions are rejected. With 1NN OT (Time), the estimated effect on the Other share is negative, -0.293 percentage points, with a 95% percentile bootstrap interval of $[-0.483, -0.046]$. For the Other share, the CoDiD of Boussim (2026), GDiD, and TDiD give positive estimated effects with intervals above zero, so their direction differs from that of 1NN OT (Time).

2. Setup and notation

Section 2.1 fixes the probability notation and the manifold-valued potential outcomes, and Section 2.2 collects the manifold operations and the common chart used throughout the paper. The important point is that these geometric objects are local. Logarithmic maps, parallel transports, Fréchet means, and common-chart coordinates are used only on regions where they are uniquely defined and stable.

2.1. **Setup.** In what follows, we assume that all random elements are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The expectation operator $\mathbb{E}$ is taken with respect to $\mathbb{P}$. For any random element X, we write $\mathbb{P} \circ X^{-1}$ for its law. For measurable spaces E and F, a measurable map $S : E \to F$, and a probability measure λ on E, let $S_{\#}\lambda$ denote the pushforward of λ by S. That is, for any measurable set $A \subset F$, $S_{\#}\lambda(A) := \lambda(S^{-1}(A))$.

Let $\mathcal{M}$ be a d-dimensional smooth manifold equipped with a Riemannian metric $g_{\mathcal{M}} = \{g_{\mathcal{M},p} : p \in \mathcal{M}\}$. For each $p \in \mathcal{M}$, $g_{\mathcal{M},p}$ is an inner product on the tangent space $T_p\mathcal{M}$, varying smoothly with p. Let $d_{\mathcal{M}}$ denote the geodesic distance induced by $g_{\mathcal{M}}$. The norm of a tangent vector $v \in T_p\mathcal{M}$ is $\|v\|_p := \{g_{\mathcal{M},p}(v, v)\}^{1/2}$. When the base point is clear from the context, we simply write $\|v\|$.

Let $G_i \in \{T, C\}$ indicate the group membership of individual i, where T and C represent the treatment and control groups, respectively. In our baseline difference-in-differences setting, $t = 0$ corresponds to the pre-treatment period, and $t = 1$ corresponds to the post-treatment period. Let $O_{i,t}(0), O_{i,t}(1) \in \mathcal{M}$ denote the manifold-valued potential outcomes. The observed outcome is $O_{i,t} = O_{i,t}(0)$ if $G_i = C$ or $t = 0$, and $O_{i,t} = O_{i,t}(1)$ if $G_i = T$ and $t = 1$. For group $a \in \{T, C\}$,

time t, and potential treatment state $z \in \{0,1\}$, let $O_{a,t}(z)$ denote a generic random outcome that follows the conditional distribution of $O_{i,t}(z)$ given $G_i = a$. Consequently, the observed generic outcomes are $O_{C,0} = O_{C,0}(0)$, $O_{C,1} = O_{C,1}(0)$, $O_{T,0} = O_{T,0}(0)$, and $O_{T,1} = O_{T,1}(1)$. The untreated counterfactual outcome for the treated group in the post-treatment period is denoted by $O_{T,1}(0)$.

2.2. **Manifold background.** This subsection states the condition on $\mathcal{M}$ that is maintained throughout and collects the geometric operations used by GDiD, TDiD, and MCiC.

Assumption 2.1 (Setup)**.** *The manifold $\mathcal{M}$ is connected, and the metric space $(\mathcal{M}, d_{\mathcal{M}})$ is complete.*

Here $\mathcal{M}$ is connected if it cannot be written as the union of two disjoint nonempty open subsets. For a manifold, this is equivalent to any two points being joined by a piecewise smooth curve, so the geodesic distance $d_{\mathcal{M}}(p,q)$ defined below is finite for all $p, q \in \mathcal{M}$. The metric space $(\mathcal{M}, d_{\mathcal{M}})$ is complete if every Cauchy sequence in $\mathcal{M}$ with respect to $d_{\mathcal{M}}$ converges to a point of $\mathcal{M}$. Assumption 2.1 is a regularity condition on the manifold itself. It is used through the Hopf–Rinow theorem, stated with the exponential map below, which guarantees, for example, that the exponential map at every point is defined on the whole tangent space.

Throughout this subsection, the unit sphere $\mathbb{S}^2 := \{x \in \mathbb{R}^3 : \|x\| = 1\}$ serves as a running example. Its tangent space at $p \in \mathbb{S}^2$ is the plane $T_p\mathbb{S}^2 = \{v \in \mathbb{R}^3 : \langle v, p\rangle = 0\}$, and its Riemannian metric is the restriction of the Euclidean inner product of $\mathbb{R}^3$ to these planes, so that $d = 2$. The sphere is connected and compact, and hence it satisfies Assumption 2.1. Section 6.1 represents three-category vote-share compositions on its positive orthant $\mathbb{S}^2_{++} := \mathbb{S}^2 \cap \mathbb{R}^3_{>0}$ through the square-root map.

Length, the Levi–Civita connection, and geodesics. For a piecewise smooth curve $c : [0,1] \to \mathcal{M}$, define its length by $\mathrm{length}(c) := \int_0^1 \|\dot{c}(s)\|_{c(s)}\,\mathrm{d}s$. The geodesic distance is $d_{\mathcal{M}}(p,q) := \inf \mathrm{length}(c)$, where the infimum is over piecewise smooth curves with $c(0) = p$ and $c(1) = q$. Let ∇ denote the Levi–Civita connection of $g_{\mathcal{M}}$. It is the unique affine connection that is torsion-free, $\nabla_X Y - \nabla_Y X = [X, Y]$, and compatible with the metric, $X\{g_{\mathcal{M}}(Y,Z)\} = g_{\mathcal{M}}(\nabla_X Y, Z) + g_{\mathcal{M}}(Y, \nabla_X Z)$ for smooth vector fields X, Y, and Z on $\mathcal{M}$. For a smooth vector field V along a smooth curve c, write $\nabla_{\dot{c}(s)} V(s)$ for its covariant derivative along c. A smooth curve γ is a geodesic if $\nabla_{\dot{\gamma}(s)}\dot{\gamma}(s) = 0$ for every s, that is, if its acceleration within $\mathcal{M}$ vanishes. Geodesics have constant speed and locally

minimize length. A geodesic $\eta : [0,1] \to \mathcal{M}$ from p to q is called minimizing if $\mathrm{length}(\eta) = d_{\mathcal{M}}(p,q)$. These notions are standard. For a more detailed treatment, see, for example, Lee (2018).

On $\mathbb{S}^2$, the length of a curve is its Euclidean arc length in $\mathbb{R}^3$, and $d_{\mathbb{S}^2}(p,q) = \arccos\langle p, q\rangle$. The covariant derivative of a vector field V along a curve c on $\mathbb{S}^2$ is the tangential part of its ordinary derivative, $\nabla_{\dot{c}(s)}V(s) = \dot{V}(s) - \langle \dot{V}(s), c(s)\rangle c(s)$. A curve on the sphere is therefore a geodesic exactly when its acceleration in $\mathbb{R}^3$ is orthogonal to the sphere, and the geodesics are the great circles traversed at constant speed. The arcs that join the marked points in Figure 2 are such great-circle geodesics.

Exponential and logarithmic maps. The Hopf–Rinow theorem states that a connected Riemannian manifold is complete as a metric space if and only if every geodesic extends to a geodesic defined on all of $\mathbb{R}$, and that in this case every closed and bounded subset is compact and any two points are joined by a minimizing geodesic. For $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$, let γ_v be the unique geodesic satisfying $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$. Under Assumption 2.1, γ_v is defined on all of $\mathbb{R}$, so the exponential map $\mathrm{Exp}_p : T_p\mathcal{M} \to \mathcal{M}$, $\mathrm{Exp}_p(v) := \gamma_v(1)$, is defined on the entire tangent space $T_p\mathcal{M}$. A normal neighborhood of p is the image under Exp_p of a star-shaped open neighborhood of the origin in $T_p\mathcal{M}$ on which Exp_p is a diffeomorphism onto its image, where a set is star-shaped if it contains the segment from the origin to each of its points. When a point q belongs to a normal neighborhood of p such that the unique minimizing geodesic from p to q exists, we define the logarithmic map by $\mathrm{Log}_p(q) := \dot{\eta}_{p\to q}(0) \in T_p\mathcal{M}$, where $\eta_{p\to q} : [0,1] \to \mathcal{M}$ is the unique minimizing geodesic satisfying $\eta_{p\to q}(0) = p$ and $\eta_{p\to q}(1) = q$. In this case, $\mathrm{Exp}_p\{\mathrm{Log}_p(q)\} = q$ and $\|\mathrm{Log}_p(q)\|_p = d_{\mathcal{M}}(p,q)$.

On $\mathbb{S}^2$, for $p \in \mathbb{S}^2$ and a nonzero $v \in T_p\mathbb{S}^2$, $\mathrm{Exp}_p(v) = \cos(\|v\|)\, p + \sin(\|v\|)\, v/\|v\|$, which is the point reached by following the great circle through p in the direction v for arc length $\|v\|$. For $q \neq \pm p$ and $\theta := \arccos\langle p, q\rangle$, $\mathrm{Log}_p(q) = \theta\,(q - \cos\theta\, p)/\sin\theta$. This is the tangent vector at p that points along the great circle toward q and whose length equals the arc length θ. The logarithmic map is defined at every $q \neq -p$. The antipode $-p$ is reached from p by every great circle, so no unique minimizing geodesic from p to $-p$ exists.

Parallel transport. For a smooth curve $\eta : [0,1] \to \mathcal{M}$, let $\Pi_\eta : T_{\eta(0)}\mathcal{M} \to T_{\eta(1)}\mathcal{M}$ denote parallel transport along η. That is, for $v \in T_{\eta(0)}\mathcal{M}$, if V is the unique vector field along η satisfying $\nabla_{\dot{\eta}(s)}V(s) = 0$ and $V(0) = v$, then $\Pi_\eta v := V(1)$. Since the Levi–Civita connection is metric-compatible, Π_η is a linear isometry; namely, for any $u, v \in T_{\eta(0)}\mathcal{M}$,

$$g_{\mathcal{M},\eta(1)}(\Pi_\eta u, \Pi_\eta v) = g_{\mathcal{M},\eta(0)}(u, v).$$

On $\mathbb{S}^2$, parallel transport along the minimizing geodesic $\eta_{p\to q}$ from p to $q \neq -p$ has the closed form $\Pi_{\eta_{p\to q}} v = v - \{\langle v, q\rangle/(1+\langle p, q\rangle)\}(p+q)$ for $v \in T_p\mathbb{S}^2$. It is the rotation of $\mathbb{R}^3$ about the axis orthogonal to the plane of the great circle through p and q that carries p to q, restricted to $T_p\mathbb{S}^2$. The transported vector therefore keeps its length and its angle with the great circle. For example, a vector pointing north at a point on the equator still points north after transport along the equator.

Strongly convex neighborhoods and geodesic transport. An open subset $\mathcal{U} \subset \mathcal{M}$ is called strongly convex if, for every pair of points $x, y \in \mathcal{U}$, there exists a unique minimizing geodesic $\eta_{x\to y} : [0,1] \to \mathcal{M}$ from x to y, and this geodesic remains entirely in $\mathcal{U}$. That is, $\eta_{x\to y}(0) = x$, $\eta_{x\to y}(1) = y$, $\mathrm{length}(\eta_{x\to y}) = d_{\mathcal{M}}(x, y)$, and $\eta_{x\to y}([0,1]) \subset \mathcal{U}$. In particular, if $x, y \in \mathcal{U}$, then the logarithmic vector $\mathrm{Log}_x(y)$ is unambiguously defined as the initial velocity of this unique minimizing geodesic.

Whenever points $a, b, w \in \mathcal{M}$ belong to a common strongly convex neighborhood $\mathcal{U}$, define the geodesic transport map by

$$\Gamma_{a,b}(w) := \mathrm{Exp}_w\{\Pi_{\eta_{a\to w}} \mathrm{Log}_a(b)\},$$

where $\eta_{a\to w}$ is the unique minimizing geodesic from a to w. The strong convexity of $\mathcal{U}$ ensures that both $\mathrm{Log}_a(b)$ and the path $\eta_{a\to w}$ used for parallel transport are uniquely defined. Completeness of $\mathcal{M}$ then ensures, by Hopf–Rinow, that the final exponential map Exp_w is defined on all of $T_w\mathcal{M}$.

Thus, $\Gamma_{a,b}(w)$ is well-defined on this local domain. It is the point obtained by first representing the variation from a to b as the tangent vector $\mathrm{Log}_a(b) \in T_a\mathcal{M}$, then parallel transporting this vector along the unique minimizing geodesic from a to w, and finally applying the transported variation from the point w via the exponential map.

On $\mathbb{S}^2$, an open spherical cap $\{x \in \mathbb{S}^2 : d_{\mathbb{S}^2}(x, o) < r\}$ with center o and radius $r < \pi/2$ is strongly convex. For three points a, b, and w in such a cap, $\Gamma_{a,b}(w)$ is obtained by carrying the arrow $\mathrm{Log}_a(b)$ along the great circle from a to w without turning it and then walking from w along the great circle in the transported direction for the arc length $d_{\mathbb{S}^2}(a, b)$. Figure 2 draws these steps. In Euclidean space the same construction gives $\Gamma_{a,b}(w) = w + (b - a)$, whereas on the sphere the transported arrow is bent along the great circle that leaves w in its direction.

Reference point and common tangent-space chart. We fix a reference point $\mu_{\mathrm{ref}} \in \mathcal{M}$. When using tangent space coordinates at μ_{ref}, we also fix an orthonormal frame via a linear isometry $B_{\mathrm{ref}} : \mathbb{R}^d \to T_{\mu_{\mathrm{ref}}}\mathcal{M}$, where $\mathbb{R}^d$ is equipped with the standard Euclidean inner product.

Let $\mathcal{U}_{\mathrm{ref}}$ be a normal neighborhood of μ_{ref} that does not meet the cut locus of μ_{ref}, that is, the set of points where the minimizing geodesic from μ_{ref} ceases to be uniquely defined or ceases to remain

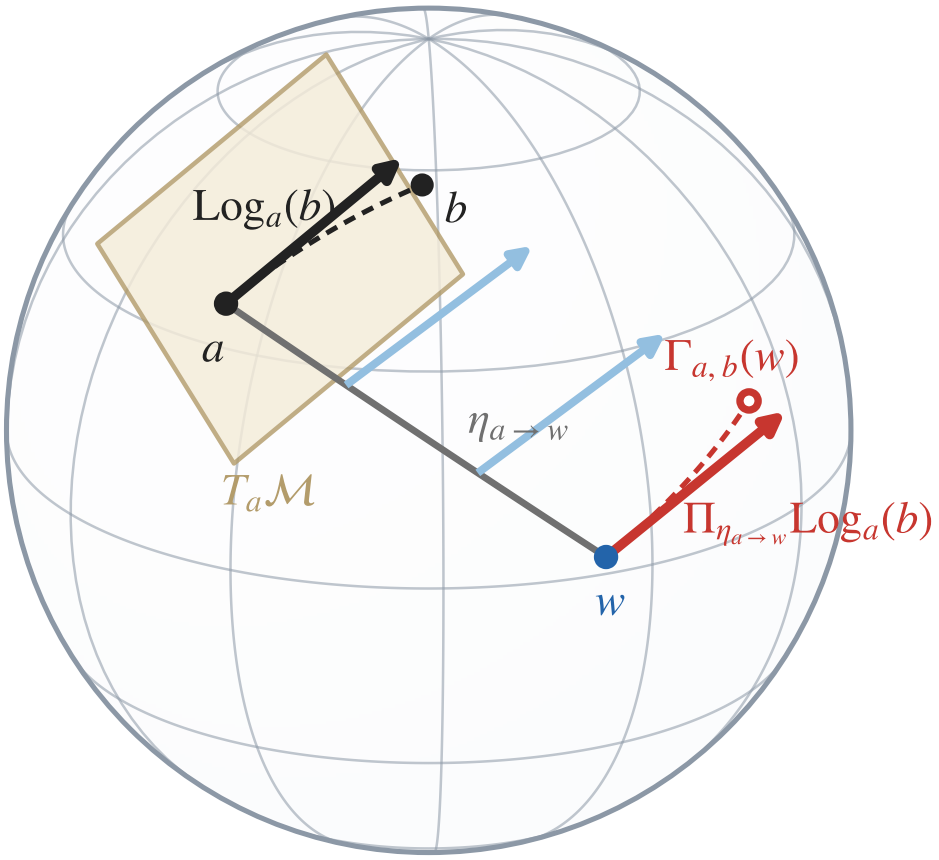


FIGURE 2. Geodesic transport on the unit sphere $\mathbb{S}^2$. The black arrow at a is $\mathrm{Log}_a(b)$, drawn in the tangent plane $T_a\mathcal{M}$, and the dashed black arc is the minimizing geodesic from a to b. The light blue arrows are the parallel transports of $\mathrm{Log}_a(b)$ to intermediate points of the minimizing geodesic $\eta_{a\to w}$ (gray). The red arrow at w is $\Pi_{\eta_{a\to w}}\mathrm{Log}_a(b)$, and the dashed red arc ends at $\Gamma_{a,b}(w) = \mathrm{Exp}_w\{\Pi_{\eta_{a\to w}}\mathrm{Log}_a(b)\}$.

minimizing. Then $\mathrm{Log}_{\mu_{\mathrm{ref}}}$ on $\mathcal{U}_{\mathrm{ref}}$ is the inverse of the restriction of $\mathrm{Exp}_{\mu_{\mathrm{ref}}}$ to the star-shaped neighborhood of the origin in $T_{\mu_{\mathrm{ref}}}\mathcal{M}$ that defines this normal neighborhood. On $\mathcal{U}_{\mathrm{ref}}$, define the common tangent space chart by

$$\Psi_{\mathrm{ref}}(x) := B_{\mathrm{ref}}^{-1}\,\mathrm{Log}_{\mu_{\mathrm{ref}}}(x), \qquad x \in \mathcal{U}_{\mathrm{ref}}.$$

Its inverse on the chart image is $\Psi_{\mathrm{ref}}^{-1}(y) = \mathrm{Exp}_{\mu_{\mathrm{ref}}}(B_{\mathrm{ref}}y)$. Thus $\mathrm{Log}_{\mu_{\mathrm{ref}}}$ and $\mathrm{Exp}_{\mu_{\mathrm{ref}}}$ put the points of $\mathcal{U}_{\mathrm{ref}}$ in one-to-one correspondence with the tangent vectors in that star-shaped neighborhood of

the origin, and B_{ref}^{-1} records each of these vectors by its d coordinates. A point $x \in \mathcal{U}_{\mathrm{ref}}$ corresponds to the tangent vector $\mathrm{Log}_{\mu_{\mathrm{ref}}}(x)$, whose length is $d_{\mathcal{M}}(\mu_{\mathrm{ref}}, x)$ and whose direction is the initial direction of the minimizing geodesic from μ_{ref} to x. The correspondence is local. On a larger set of tangent vectors, $\mathrm{Exp}_{\mu_{\mathrm{ref}}}$ can map different vectors to the same point.

On $\mathbb{S}^2_{++}$, a natural reference point for compositions is $\mu_{\mathrm{ref}} = (1,1,1)^\top/\sqrt{3}$, the square root of the uniform composition. The cut locus of μ_{ref} is its antipode $-\mu_{\mathrm{ref}}$, which lies outside the positive orthant, so $\mathbb{S}^2_{++}$ is contained in the normal neighborhood $\mathbb{S}^2 \setminus \{-\mu_{\mathrm{ref}}\}$ and the chart Ψ_{ref} is defined on all compositions. Figure 3 shows the chart. Each point of $\mathbb{S}^2_{++}$ is sent to the point of the tangent plane at μ_{ref} that has the same distance from μ_{ref} and lies in the same direction, and B_{ref}^{-1} reads off its two coordinates. Conversely, $\mathrm{Exp}_{\mu_{\mathrm{ref}}}$ wraps each of these points of the tangent plane back onto the sphere at the original point, so each composition corresponds to exactly one chart point. The chart keeps distances and directions measured from μ_{ref} but distorts distances between other points, and the distortion grows with the distance from μ_{ref}.

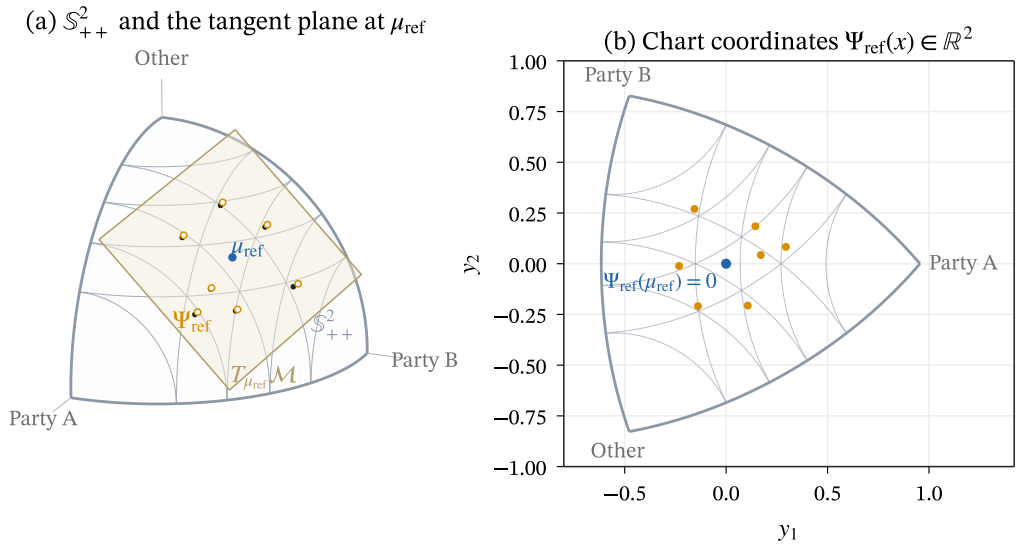


FIGURE 3. The common tangent-space chart for three-category compositions. Panel (a) shows the positive orthant $\mathbb{S}^2_{++}$ of the unit sphere, the images of the lines on which one share is constant (thin gray curves), and the tangent plane $T_{\mu_{\mathrm{ref}}}\mathcal{M}$ at $\mu_{\mathrm{ref}} = (1,1,1)^\top/\sqrt{3}$. Each black point is the square root of a composition, and the orange segment joins it to its image $\mu_{\mathrm{ref}} + \mathrm{Log}_{\mu_{\mathrm{ref}}}(x)$ in the tangent plane (hollow orange point). Panel (b) shows the chart coordinates $\Psi_{\mathrm{ref}}(x) \in \mathbb{R}^2$ of the same compositions and of the boundary of $\mathbb{S}^2_{++}$, with the first coordinate pointing toward the Party A vertex. Here B_{ref} maps the first and second standard basis vectors of $\mathbb{R}^2$ to the unit tangent vectors $(2,-1,-1)^\top/\sqrt{6}$ and $(0,1,-1)^\top/\sqrt{2}$ at μ_{ref}, so the second coordinate increases in the direction from the Other vertex to the Party B vertex.

3. TWO MEAN-BASED DID

This section reviews two mean-based difference-in-differences (DID) benchmarks before turning to the distributional construction of Section 5. Geodesic DID (GDiD) works with intrinsic Fréchet means and geodesic transport. Tangent-space DID (TDiD) first represents outcomes in the common tangent chart and then applies an ordinary Euclidean mean DID restriction. These benchmarks

are useful because they make clear which target is being identified and which geometric restrictions are being imposed before the paper introduces the transport-based counterfactual distribution.

3.1. **Geodesic DID.** This subsection reviews the geodesic DID (GDiD) identification argument of Zhou et al. (2026) on the Riemannian manifold $\mathcal{M}$. We impose the following restrictions on the outcomes of the three observed group–period pairs and on the treated group's untreated post-treatment outcome.

Assumption 3.1 (GDiD identification)**.** *For each observed group–period pair* $(a,t) \in \{(C,0),(C,1),(T,0)\}$*, define the population Fréchet functional by* $F_{a,t}(q) := \mathbb{E}[d_{\mathcal{M}}(O_{a,t},q)^2]$ *for* $q \in \mathcal{M}$*. Similarly, define the untreated counterfactual Fréchet functional for the treated group in the post-treatment period by* $F^{(0)}_{T,1}(q) := \mathbb{E}[d_{\mathcal{M}}\{O_{T,1}(0),q\}^2]$ *for* $q \in \mathcal{M}$*. The following restrictions hold.*

(1) *The Fréchet functionals* $F_{C,0}$*,* $F_{C,1}$*,* $F_{T,0}$*, and* $F^{(0)}_{T,1}$ *are finite and have unique minimizers. Denote these minimizers by* $\mu_{C,0}$*,* $\mu_{C,1}$*,* $\mu_{T,0}$*, and* $\mu_{T,1}(0)$*, respectively.*

(2) *There exists an open strongly convex neighborhood* $\mathcal{U}_{\mathrm{GDiD}} \subset \mathcal{M}$ *such that* $\mu_{C,0}$*,* $\mu_{C,1}$*,* $\mu_{T,0}$*,* $\mu_{T,1}(0) \in \mathcal{U}_{\mathrm{GDiD}}$*.*

(3) *Let* $\eta_{C\to T}$ *denote the unique minimizing geodesic from* $\mu_{C,0}$ *to* $\mu_{T,0}$*. The geodesic parallel trends condition holds:* $\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\} = \Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})$*. Consequently, using the geodesic transport notation* Γ *introduced in Section 2.2,* $\Gamma_{\mu_{C,0},\mu_{C,1}}(\mu_{T,0}) = \mu_{T,1}(0)$*.*

Condition (1) makes the four Fréchet means well-defined targets. Uniqueness is not automatic on a nonlinear manifold. On the unit circle with its intrinsic distance, for example, a law that puts probability one half on each of two antipodal points has two Fréchet means, one at the midpoint of each arc between them, so condition (1) fails for a group–period pair whose outcome has this law. Condition (2) is a convenient sufficient condition. It guarantees in one step that the logarithmic vectors and the minimizing geodesics used by GDiD are uniquely defined. The identification argument itself needs only $\mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})$, $\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\}$, and $\eta_{C\to T}$ to be uniquely defined. Condition (3) is the identifying restriction. It requires the treated group's untreated mean movement, written as a tangent vector at $\mu_{T,0}$, to equal the control group's mean movement after parallel transport along $\eta_{C\to T}$. When $\mathcal{M} = \mathbb{R}^d$ with the Euclidean metric, and in particular for a scalar outcome with $d = 1$, the Fréchet means are the expectations, $\mathrm{Log}_p(q) = q - p$, and parallel transport is the identity. Condition (3) then reads $\mu_{T,1}(0) - \mu_{T,0} = \mu_{C,1} - \mu_{C,0}$, which is the usual parallel trends assumption on means. Figure 4 illustrates the construction on $\mathbb{S}^2$.

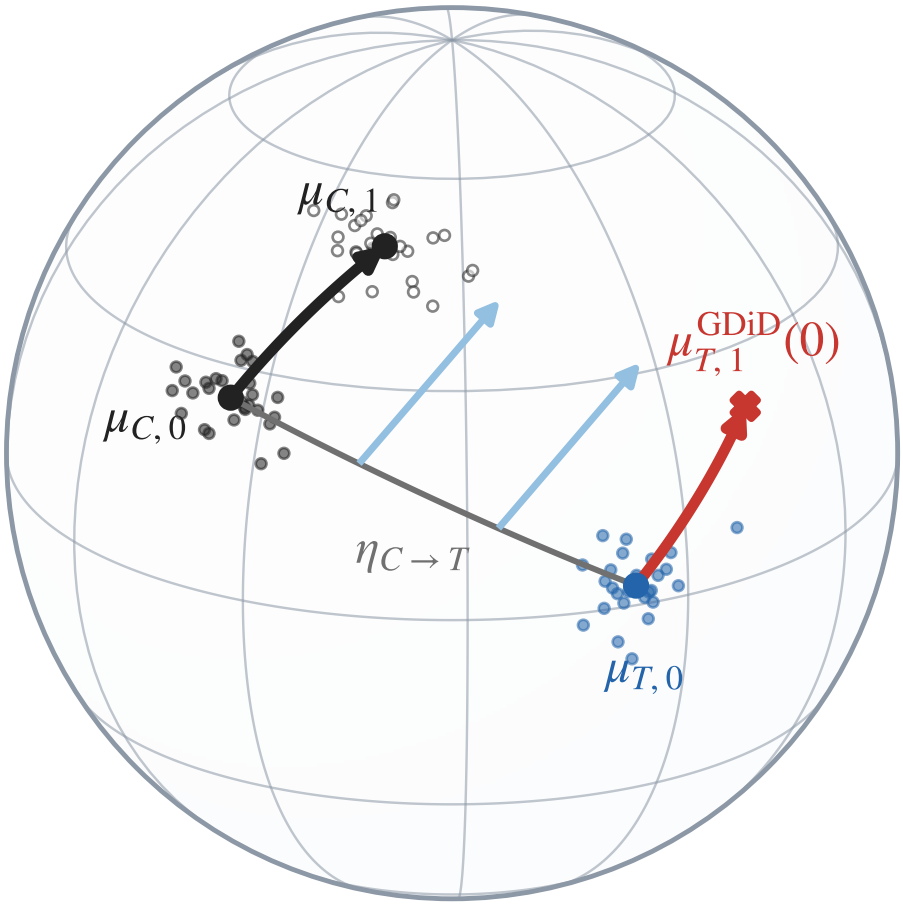


FIGURE 4. The GDiD construction on the unit sphere $\mathbb{S}^2$. Small points are sampled outcomes for the group–period pairs $(C,0)$ (black), $(C,1)$ (hollow black), and $(T,0)$ (blue), and large points are their sample Fréchet means. The black arc is the control mean movement from $\mu_{C,0}$ to $\mu_{C,1}$. The light blue arrows are parallel transports of $\mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})$ along the minimizing geodesic $\eta_{C\to T}$ (gray), and the red arc applies the transported vector at $\mu_{T,0}$, ending at $\mu_{T,1}^{\mathrm{GDiD}}(0) = \Gamma_{\mu_{C,0},\mu_{C,1}}(\mu_{T,0})$. The samples are synthetic.

Under these restrictions, the GDiD counterfactual mean is identified from the outcome laws of the three observed group–period pairs. The following theorem adapts the identification argument of Theorem 1 of Zhou et al. (2026) to the counterfactual Fréchet mean on $\mathcal{M}$.

Theorem 3.1 (GDiD identification)**.** *Suppose that Assumptions 2.1 and 3.1 hold. Then the untreated counterfactual Fréchet mean for the treated group in the post-treatment period is identified as* $\mu_{T,1}^{\mathrm{GDiD}}(0) := \mathrm{Exp}_{\mu_{T,0}}\left(\Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})\right)$*. Equivalently,* $\mu_{T,1}^{\mathrm{GDiD}}(0) = \Gamma_{\mu_{C,0},\mu_{C,1}}(\mu_{T,0})$*. Under the geodesic parallel trends condition in Assumption 3.1 (3), this identified quantity equals the true untreated counterfactual Fréchet mean:* $\mu_{T,1}^{\mathrm{GDiD}}(0) = \mu_{T,1}(0)$*.*

Theorem 3.1 identifies a mean object, not the full untreated counterfactual distribution. It says that if the control group's Fréchet-mean movement can be transported to the treated group's baseline mean by parallel transport along the local geodesic, then the transported point is the treated group's untreated post-period Fréchet mean. Thus the GDiD target is the curved-space analogue of adding the control group's mean change to the treated group's baseline mean, with ordinary subtraction and addition replaced by Log, parallel transport, and Exp. The proof is given in Appendix A.1.

For estimation, we use the sample Fréchet mean for each observed group–period pair. For $(a,t) \in \{(C,0),(C,1),(T,0)\}$, let

$$\widehat{\mu}_{a,t} \in \operatorname*{arg\,min}_{p\in\mathcal{M}} \frac{1}{n_a} \sum_{i:G_i=a} d_{\mathcal{M}}(O_{i,t},p)^2, \qquad n_a := \sum_{i=1}^{n} \mathbf{1}\{G_i = a\}.$$

If $\widehat{\eta}_{C\to T}$ denotes the unique minimizing geodesic from $\widehat{\mu}_{C,0}$ to $\widehat{\mu}_{T,0}$, then the sample version of the GDiD counterfactual mean is given by

$$\widehat{\mu}_{T,1}^{\mathrm{GDiD}}(0) := \mathrm{Exp}_{\widehat{\mu}_{T,0}}\left(\Pi_{\widehat{\eta}_{C\to T}} \mathrm{Log}_{\widehat{\mu}_{C,0}}(\widehat{\mu}_{C,1})\right).$$

3.2. **Tangent DID.** This subsection formulates a tangent-space DID (TDiD) identification argument using the common tangent-space chart Ψ_{ref} introduced in the setup. Unlike GDiD, which operates on Fréchet means and geodesic transport, TDiD first maps outcomes into the fixed Euclidean chart at μ_{ref}, applies the usual additive DID restriction in that chart, and then maps the resulting chart mean back to the manifold.

We now state the TDiD restriction in the coordinates of the common chart. For notational convenience, write $P_{C,0} := \mathbb{P} \circ O_{C,0}^{-1}$, $P_{C,1} := \mathbb{P} \circ O_{C,1}^{-1}$, $P_{T,0} := \mathbb{P} \circ O_{T,0}^{-1}$, and $P_{T,1}^{(0)} := \mathbb{P} \circ \{O_{T,1}(0)\}^{-1}$ for the corresponding manifold-valued laws.

Assumption 3.2 (TDiD identification)**.** *There exist an open neighborhood* $\mathcal{U}_{\mathrm{TDiD}} \subset \mathcal{U}_{\mathrm{ref}}$ *and a compact set* $\mathcal{K}_{\mathrm{TDiD}} \subset \mathcal{U}_{\mathrm{TDiD}}$ *such that* $\mathrm{supp}(P_{C,0}) \cup \mathrm{supp}(P_{C,1}) \cup \mathrm{supp}(P_{T,0}) \cup \mathrm{supp}(P_{T,1}^{(0)}) \subset \mathcal{K}_{\mathrm{TDiD}}$*. Let* $\mathcal{Y}_{\mathrm{TDiD}} := \Psi_{\mathrm{ref}}(\mathcal{U}_{\mathrm{TDiD}})$*. Define the common-chart coordinates by* $Y_{C,0} := \Psi_{\mathrm{ref}}(O_{C,0})$*,*

$Y_{C,1} := \Psi_{\mathrm{ref}}(O_{C,1})$, $Y_{T,0} := \Psi_{\mathrm{ref}}(O_{T,0})$, *and* $Y_{T,1}(0) := \Psi_{\mathrm{ref}}\{O_{T,1}(0)\}$. *Since* $\mathcal{K}_{\mathrm{TDiD}}$ *is compact and* Ψ_{ref} *is continuous on* $\mathcal{U}_{\mathrm{TDiD}}$, *these random vectors have finite first moments. Let* $m^Y_{C,0} := \mathbb{E}[Y_{C,0}]$, $m^Y_{C,1} := \mathbb{E}[Y_{C,1}]$, $m^Y_{T,0} := \mathbb{E}[Y_{T,0}]$, *and* $m^{Y,(0)}_{T,1} := \mathbb{E}[Y_{T,1}(0)]$. *The following conditions hold.*

(1) *The tangent-space parallel trends condition holds:* $\mathbb{E}[Y_{T,1}(0) - Y_{T,0}] = \mathbb{E}[Y_{C,1} - Y_{C,0}]$.
(2) *The identified counterfactual chart mean belongs to the chart image:* $m^Y_{T,0} + \left(m^Y_{C,1} - m^Y_{C,0}\right) \in \mathcal{Y}_{\mathrm{TDiD}}$.

The support condition places the four outcome laws in a compact subset of the chart domain, which is why the chart coordinates have finite first moments. Condition (1) is the usual parallel trends restriction applied to these coordinates. It requires the treated group's expected untreated chart increment to equal the control group's expected chart increment. Condition (2) ensures that the identified chart mean can be mapped back to $\mathcal{M}$ by Ψ^{-1}_{ref}.

Under these conditions, the counterfactual chart mean and its image on $\mathcal{M}$ are identified as follows.

Theorem 3.2 (TDiD identification)**.** *Suppose that Assumptions 2.1 and 3.2 hold. Then the untreated counterfactual chart mean for the treated group in the post-treatment period is identified as* $m^{Y,(0)}_{T,1} = m^Y_{T,0} + \left(m^Y_{C,1} - m^Y_{C,0}\right)$. *Consequently, the corresponding manifold-valued chart-mean summary is identified as* $\theta^{\mathrm{TDiD}}_{T,1}(0) := \Psi^{-1}_{\mathrm{ref}}\left\{m^Y_{T,0} + \left(m^Y_{C,1} - m^Y_{C,0}\right)\right\}$. *Equivalently,* $\theta^{\mathrm{TDiD}}_{T,1}(0) = \mathrm{Exp}_{\mu_{\mathrm{ref}}}\left[B_{\mathrm{ref}}\left\{m^Y_{T,0} + \left(m^Y_{C,1} - m^Y_{C,0}\right)\right\}\right]$.

Theorem 3.2 is the ordinary DID formula applied to chart means. The control group's mean chart increment is added to the treated group's baseline chart mean, and the result is mapped back to the manifold by the inverse chart. The resulting $\theta^{\mathrm{TDiD}}_{T,1}(0)$ is a summary defined through the chart, and Remark 3.1 explains how it differs from a Fréchet mean. The proof is given in Appendix A.2. Figure 5 draws the construction in the chart used for three-category compositions.

For estimation, map the observed outcomes of the group–period pairs $(a,t) \in \{(C,0),(C,1),(T,0)\}$ to the common tangent-space chart by setting $Y_{i,t} := \Psi_{\mathrm{ref}}(O_{i,t})$. Let $n_a := \sum_{i=1}^n \mathbf{1}\{G_i = a\}$ for $a \in \{T,C\}$, and define the sample chart means by $\bar{Y}_{a,t} := n_a^{-1} \sum_{i:G_i=a} Y_{i,t}$. The sample TDiD counterfactual chart mean is

$$\widehat{m}^{Y,\mathrm{TDiD}}_{T,1}(0) := \bar{Y}_{T,0} + (\bar{Y}_{C,1} - \bar{Y}_{C,0}).$$

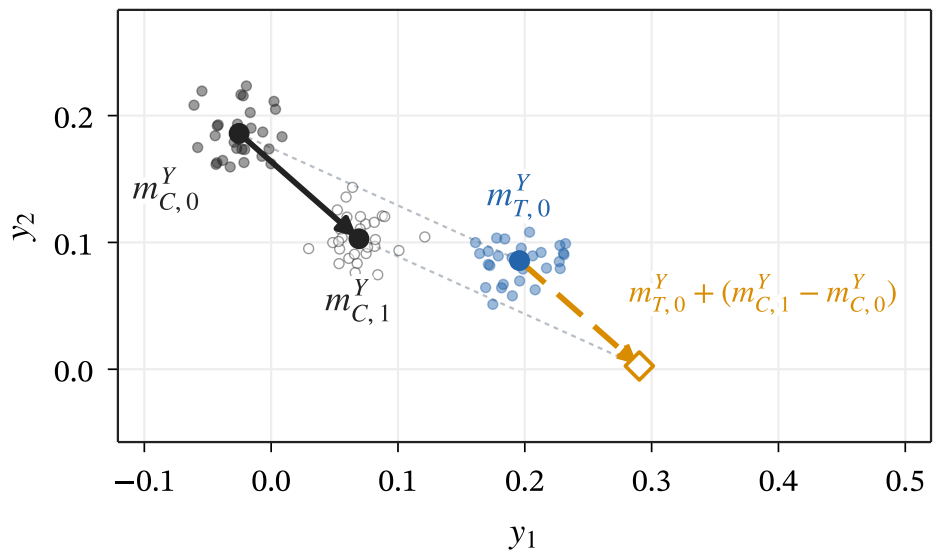


FIGURE 5. The TDiD construction in the common tangent-space chart of Figure 3. Small points are chart coordinates $\Psi_{\text{ref}}(x)$ of synthetic compositions for the group–period pairs $(C,0)$ (black), $(C,1)$ (hollow black), and $(T,0)$ (blue), and large points are their chart means. The black arrow is the control chart-mean increment $m^Y_{C,1} - m^Y_{C,0}$, and the dashed orange arrow adds the same increment to $m^Y_{T,0}$, ending at the counterfactual chart mean $m^Y_{T,0} + (m^Y_{C,1} - m^Y_{C,0})$ (hollow diamond). The dotted lines complete the parallelogram. Theorem 3.2 maps this point back to the manifold by Ψ^{-1}_{ref}.

The corresponding sample manifold-valued chart-mean summary is

$$\widehat{\theta}^{\text{TDiD}}_{T,1}(0) := \Psi^{-1}_{\text{ref}} \left\{ \widehat{m}^{Y,\text{TDiD}}_{T,1}(0) \right\}.$$

Remark 3.1 (Difference between GDiD and TDiD)**.** GDiD and TDiD differ in how the mean is taken and in the operation that adds the change of the control group to that mean. The representative point $\mu^{\text{GDiD}}_{T,1}(0)$ of GDiD is constructed from the Fréchet mean of each group–period pair by transport along geodesics. The representative point $\theta^{\text{TDiD}}_{T,1}(0)$ of TDiD is constructed by mapping the outcomes to the common chart Ψ_{ref}, taking their means, and mapping the coordinates obtained by the DID addition back by the inverse chart. To examine the condition under which the two coincide, suppose that the three Fréchet means $\mu_{C,0}$, $\mu_{C,1}$, and $\mu_{T,0}$ and both representative

points belong to the domain of the injective common chart Ψ_{ref}. For $(a,t) \in \{(C,0),(C,1),(T,0)\}$, denote the chart means $\mathbb{E}\{\Psi_{\mathrm{ref}}(O_{a,t})\}$ of the outcomes $O_{a,t}$ by $m^Y_{C,0}$, $m^Y_{C,1}$, and $m^Y_{T,0}$, respectively. Since $\Psi_{\mathrm{ref}}\{\theta^{\mathrm{TDiD}}_{T,1}(0)\} = m^Y_{T,0} + (m^Y_{C,1} - m^Y_{C,0})$ by the definition of TDiD, adding and subtracting the DID sum of the chart coordinates of the Fréchet means gives

$$\Psi_{\mathrm{ref}}\{\mu^{\mathrm{GDiD}}_{T,1}(0)\} - \Psi_{\mathrm{ref}}\{\theta^{\mathrm{TDiD}}_{T,1}(0)\} = \underbrace{\Psi_{\mathrm{ref}}\{\mu^{\mathrm{GDiD}}_{T,1}(0)\} - [\Psi_{\mathrm{ref}}(\mu_{T,0}) + \{\Psi_{\mathrm{ref}}(\mu_{C,1}) - \Psi_{\mathrm{ref}}(\mu_{C,0})\}]}_{\Delta_{\mathrm{geo}}}$$
$$- \underbrace{\left[\{m^Y_{T,0} - \Psi_{\mathrm{ref}}(\mu_{T,0})\} + \{m^Y_{C,1} - \Psi_{\mathrm{ref}}(\mu_{C,1})\} - \{m^Y_{C,0} - \Psi_{\mathrm{ref}}(\mu_{C,0})\}\right]}_{\Delta_{\mathrm{mean}}}.$$

Δ_{geo} is the difference between the operation that transports the Fréchet means along geodesics and the operation that adds their coordinates by DID. Δ_{mean} combines, with the DID signs over the three group–period pairs, the differences between the mean taken after mapping to the chart and the point to which the Fréchet mean is mapped in the chart. By the injectivity of the common chart, $\mu^{\mathrm{GDiD}}_{T,1}(0) = \theta^{\mathrm{TDiD}}_{T,1}(0)$ holds if and only if $\Delta_{\mathrm{geo}} = \Delta_{\mathrm{mean}}$. Therefore, to treat GDiD and TDiD as the same representative point, this coincidence condition needs to be checked separately. Figure 6 shows a special configuration in which the two coincide, and Figure 7 shows a configuration in which they do not.

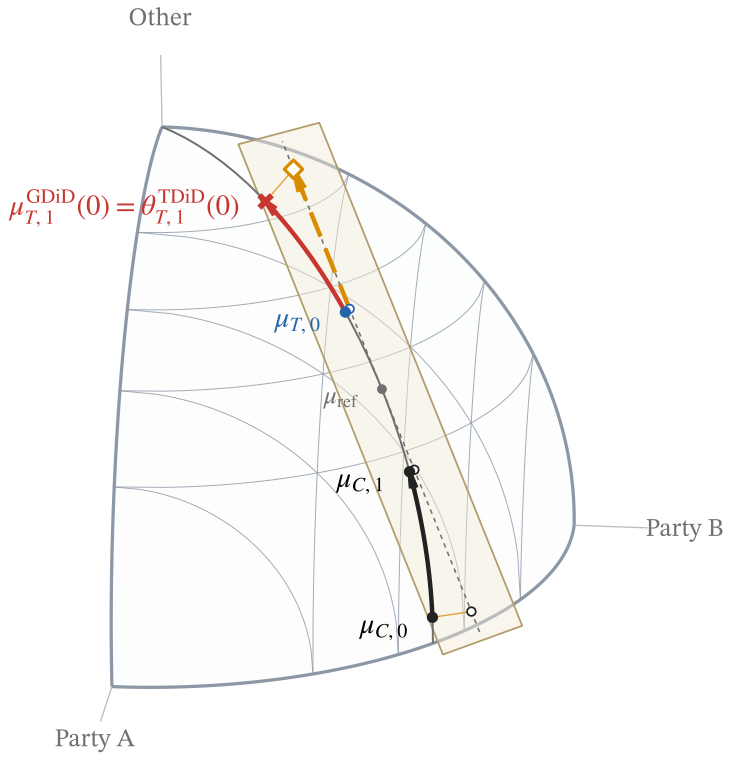


FIGURE 6. Fréchet means on one great circle through $\mu_{\mathrm{ref}} = (1,1,1)^\top/\sqrt{3}$ on the positive orthant $\mathbb{S}^2_{++}$ of Figure 3. Each outcome law is a point mass at its Fréchet mean, so $\Delta_{\mathrm{mean}} = 0$. The means $\mu_{C,0}$ and $\mu_{C,1}$ (black) and $\mu_{T,0}$ (blue) lie on the gray great circle through μ_{ref} (gray point) and the Other vertex. The black arc is the control mean movement, and the red arc applies its parallel transport at $\mu_{T,0}$ and ends at $\mu_{T,1}^{\mathrm{GDiD}}(0)$ (cross). The strip is part of the tangent plane $T_{\mu_{\mathrm{ref}}}\mathcal{M}$, in which the dotted line is the chart image of the great circle, and each thin orange segment joins a point x on the sphere to its image $\mu_{\mathrm{ref}} + \mathrm{Log}_{\mu_{\mathrm{ref}}}(x)$ (hollow points). The dashed orange arrow adds the control chart increment at the image of $\mu_{T,0}$ and ends at the image of $\theta_{T,1}^{\mathrm{TDiD}}(0)$ (hollow diamond). The thin orange segment that starts at the cross ends at this diamond, so the two counterfactuals coincide and $\Delta_{\mathrm{geo}} = 0$.

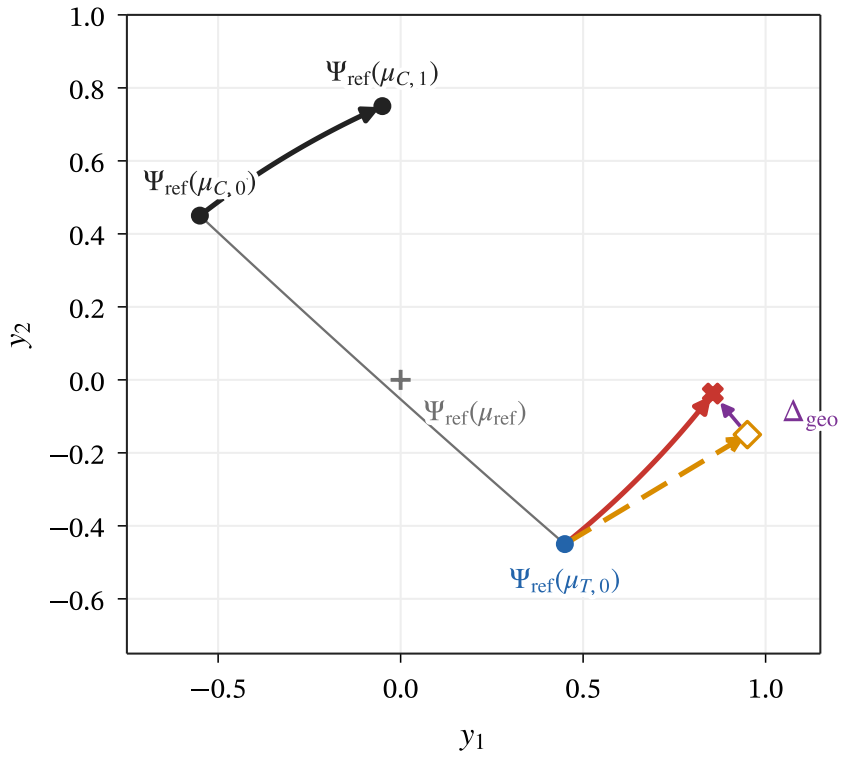


FIGURE 7. The geometric term Δ_{geo} for Fréchet means in general position on the unit sphere $\mathbb{S}^2$, drawn in the chart at $\mu_{\mathrm{ref}} = (0,0,1)^\top$. Each outcome law is a point mass at its Fréchet mean, so $\Delta_{\mathrm{mean}} = 0$. Black points are $\Psi_{\mathrm{ref}}(\mu_{C,0})$ and $\Psi_{\mathrm{ref}}(\mu_{C,1})$, the blue point is $\Psi_{\mathrm{ref}}(\mu_{T,0})$, and the gray curve is the charted minimizing geodesic $\eta_{C\to T}$. The red curve ends at the charted GDiD counterfactual $\Psi_{\mathrm{ref}}\{\mu_{T,1}^{\mathrm{GDiD}}(0)\}$ (cross), and the dashed orange arrow ends at $\Psi_{\mathrm{ref}}(\mu_{T,0}) + \{\Psi_{\mathrm{ref}}(\mu_{C,1}) - \Psi_{\mathrm{ref}}(\mu_{C,0})\}$ (hollow diamond). The purple arrow shows $\Delta_{\mathrm{geo}} \neq 0$.

4. PRETREND TESTS

This section formulates placebo pretrend tests using two pre-treatment periods $s_0 < s_1$. The two tests below are diagnostics for the mean restrictions used by GDiD and TDiD. The GDiD test asks whether the treated and control pre-period movements agree after comparing Fréchet means through geodesic transport. The TDiD test asks whether the treated and control pre-period mean increments agree in the chosen common chart. A rejection should be interpreted as evidence against the corresponding mean DID benchmark in that application.

Both tests use the following sampling assumption for the two pre-treatment periods. For the pretrend tests, write the observed outcome in group-specific notation as $O_{i,a,s_j} \in \mathcal{M}$, $a \in \{T, C\}$, $j \in \{0, 1\}$, where $i = 1, \ldots, n_a$ indexes individuals in group a. This notation represents the sample from the conditional distribution of the pre-treatment outcome given group membership. Since both s_0 and s_1 are pre-treatment periods, $O_{i,a,s_j} = O_{i,a,s_j}(0)$ a.s., $a \in \{T, C\}$, $j \in \{0, 1\}$. Let O_{a,s_j} denote a generic random outcome with the same distribution as O_{i,a,s_j}.

Assumption 4.1 (Pretrend sampling)**.** *For each group $a \in \{T, C\}$, the paired pre-treatment trajectories $\mathbf{O}_{i,a} := (O_{i,a,s_0}, O_{i,a,s_1}) \in \mathcal{M}^2$, $i = 1, \ldots, n_a$, are i.i.d. The treated and control samples are independent. Moreover, $n_a/n \to \rho_a \in (0, 1)$ for $a \in \{T, C\}$, where $n := n_T + n_C$.*

Assumption 4.1 treats each unit's pair of pre-treatment outcomes as one observation, so the two outcomes of the same unit may be dependent. The GDiD test accounts for this dependence through the joint limit of the two pre-period Fréchet means, and the TDiD test through unit-level increments. Independence between the treated and control samples and the convergence of each group's sample share determine how the two groups' sampling errors combine in the limit.

4.1. **GDiD pretrend test.** The GDiD pretrend test is based on the four pre-period Fréchet means $\mu_{T,s_0}, \mu_{T,s_1}, \mu_{C,s_0}, \mu_{C,s_1}$. It tests whether the treated group's pre-period mean trend agrees with the control group's pre-period mean trend after the latter is transported to the treated group's baseline mean.

Before stating the regularity condition, we make the geometric notation explicit. For $x \in \mathcal{M}$ and $r > 0$, the closed geodesic ball is $\overline{B}_r(x) := \{q \in \mathcal{M} : d_{\mathcal{M}}(q, x) \leq r\}$. The cut locus $C(x)$ is the set of points where the minimizing geodesic from x ceases to be uniquely defined or ceases to remain minimizing. The quantity $\mathrm{Sec}_p(\sigma)$ is the sectional curvature at p of the two-dimensional tangent plane $\sigma \subset T_p\mathcal{M}$. It measures the local curvature of the two-dimensional surface generated by geodesics tangent to σ. On the unit sphere $\mathbb{S}^2$, the closed geodesic ball $\overline{B}_r(x)$ is the spherical cap of angular radius r around x, the cut locus of x is its antipode, $C(x) = \{-x\}$, and every sectional curvature equals one. Figure 8 draws these objects. Every great circle through x reaches $-x$ after arc length π, so the minimizing geodesic from x stops being unique exactly at $-x$.

In addition to the sampling in Assumption 4.1, the GDiD test uses the following conditions on the four pre-period outcome laws and their Fréchet means, which concern the uniqueness of the means, a common chart around them, the supports of the laws, the smoothness of the squared

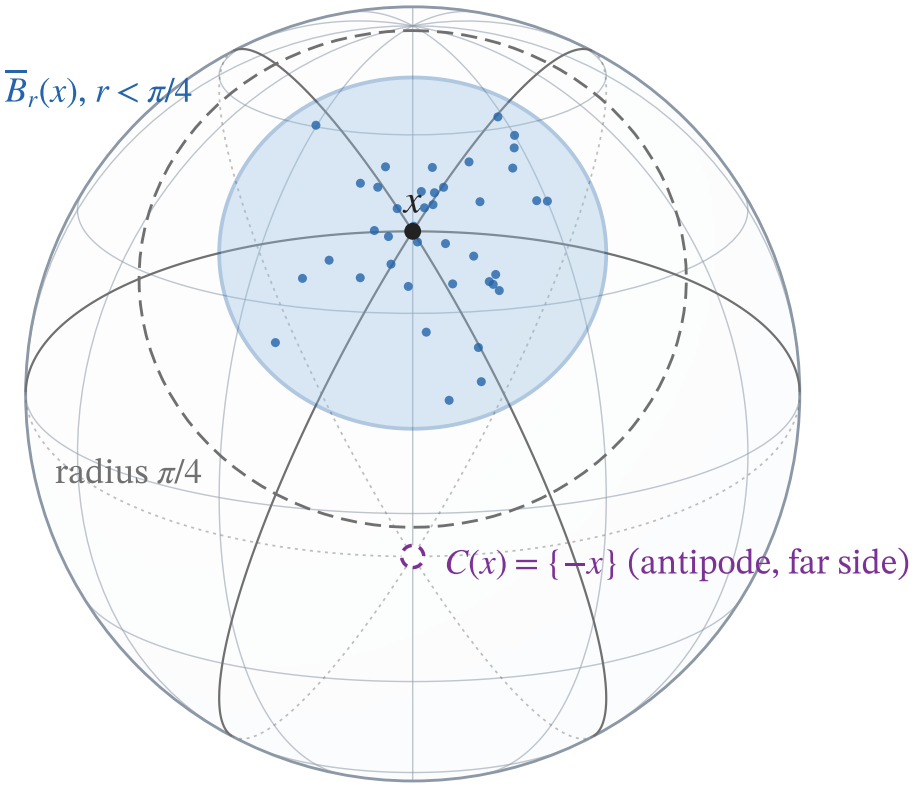


FIGURE 8. A closed geodesic ball and the cut locus on the unit sphere $\mathbb{S}^2$. The gray great circles leave x in several directions and meet again at the antipode $-x$, which lies on the far side of the sphere (dotted parts). The antipode is the cut locus $C(x)$. The shaded cap is a closed geodesic ball $\overline{B}_r(x)$ with $r = \pi/6$, which contains the support of an outcome law (blue points) and does not meet $C(x)$. The dashed circle marks the radius bound $\pi/4$ of Assumption 4.2(4) for $K = 1$.

distance in that chart, and a positive definite Hessian of each charted Fréchet functional at its minimizer.

Assumption 4.2 (Pretrend GDiD regularity)**.** *For each $a \in \{T, C\}$ and $j \in \{0, 1\}$, let $P_{a,j}$ denote the law of O_{a,s_j} and define $F_{a,j}(q) := \mathbb{E}[d_{\mathcal{M}}(O_{a,s_j}, q)^2]$ for $q \in \mathcal{M}$.*

(1) *The Fréchet functional $F_{a,j}$ is finite and has a unique minimizer μ_{a,s_j}.*

(2) *There exists an open strongly convex coordinate neighborhood* $\mathcal{U}_{\mathrm{PG}} \subset \mathcal{M}$ *such that* μ_{ref}, μ_{T,s_0}, μ_{T,s_1}, μ_{C,s_0}, *and* μ_{C,s_1} *belong to* $\mathcal{U}_{\mathrm{PG}}$. *Let* $\Psi_{\mathrm{PG}} : \mathcal{U}_{\mathrm{PG}} \to \mathbb{R}^d$ *be a smooth chart.*

(3) *There exists a compact set* $\mathcal{K}_{\mathrm{PG}} \subset \mathcal{U}_{\mathrm{PG}}$ *such that* $P_{a,j}(\mathcal{K}_{\mathrm{PG}}) = 1$ *for all* $a \in \{T, C\}$ *and* $j \in \{0, 1\}$.

(4) *For each* $a \in \{T, C\}$ *and* $j \in \{0, 1\}$, *there exist* $x_{a,j} \in \mathcal{U}_{\mathrm{PG}}$ *and* $r_{a,j} > 0$ *such that, with* $B_{a,j} := \overline{B}_{r_{a,j}}(x_{a,j})$, *one has* $\mathrm{supp}(P_{a,j}) \subset B_{a,j} \subset \mathcal{U}_{\mathrm{PG}}$ *and* $B_{a,j} \cap C(x_{a,j}) = \emptyset$, *where* $C(x_{a,j})$ *denotes the cut locus of* $x_{a,j}$. *Let* $\overline{\kappa}_{a,j} := \sup\{\mathrm{Sec}_p(\sigma) : p \in B_{a,j}, \sigma \subset T_p\mathcal{M}$ *is a two-dimensional subspace*$\}$, *and define* $K_{a,j} := \sqrt{\overline{\kappa}_{a,j}}$ *if* $\overline{\kappa}_{a,j} > 0$ *and* $K_{a,j} := 0$ *if* $\overline{\kappa}_{a,j} \leq 0$. *The radius satisfies* $r_{a,j} < \pi/(4K_{a,j})$, *with the convention that* $\pi/(4K_{a,j}) = +\infty$ *when* $K_{a,j} = 0$.

(5) *For each* $a \in \{T, C\}$ *and* $j \in \{0, 1\}$, *there exists an open neighborhood of* $\Psi_{\mathrm{PG}}(\mu_{a,s_j})$ *in* $\Psi_{\mathrm{PG}}(\mathcal{U}_{\mathrm{PG}})$ *on which, for every* $o \in \mathrm{supp}(P_{a,j})$, *the map* $u \mapsto d_{\mathcal{M}}\{o, \Psi_{\mathrm{PG}}^{-1}(u)\}^2$ *is twice continuously differentiable. Its first and second derivatives are jointly continuous in* o *and* u *and bounded over* $o \in \mathrm{supp}(P_{a,j})$ *and* u *in this neighborhood. Moreover, the Hessian of* $u \mapsto F_{a,j}\{\Psi_{\mathrm{PG}}^{-1}(u)\}$ *at* $u = \Psi_{\mathrm{PG}}(\mu_{a,s_j})$ *is positive definite.*

Assumption 4.2 imposes local conditions around the four pre-period outcome laws and their Fréchet means rather than global convexity requirements on $\mathcal{M}$. Condition (1) makes the four pre-period Fréchet means well-defined targets, as condition (1) of Assumption 3.1 does for the four Fréchet means of Section 3.1. Assumption 4.2(2) supplies the local chart Ψ_{PG} in which these derivatives are taken, and Assumption 4.2(3) supplies the compact support used for the moment conditions. Assumption 4.2(4) is a concentration condition. It localizes each outcome distribution in a closed geodesic ball that avoids the cut locus of its center and whose radius is small relative to the curvature. This is the support condition of Theorem 2.2 of Bhattacharya and Patrangenaru (2005), whose proof uses the convexity result of Karcher (1977). The analytic regularity used by the Fréchet mean CLT is stated separately in Assumption 4.2(5), namely second-order differentiability of the squared distance near each population mean and a positive definite Hessian of the charted Fréchet functional. In Figure 8, the shaded cap is such a ball. It contains the support of the outcome law, stays away from the cut locus $-x$ of its center, and has radius $\pi/6$, below the bound $\pi/4$ that the sphere's curvature $K = 1$ gives.

The GDiD pretrend test is built from the four pre-period sample Fréchet means, and its first ingredient is a joint central limit theorem for them. For one i.i.d. sample, Theorem 2.1 of Bhattacharya and Patrangenaru (2005) shows that sample Fréchet means are consistent and asymptotically normal in a coordinate chart, under conditions that consist of a unique Fréchet mean, a coordinate chart that carries the whole law, differentiability of the squared distance in the chart, moment conditions, and nonsingularity of the Hessian of the charted Fréchet functional. Lemma 4.1 below puts this theorem into the form used here, in which the four means are treated jointly, the two pre-period outcomes of each unit may be dependent, and the limit is stated in one product chart.

Under Assumptions 4.1 and 4.2, define the sample Fréchet means as measurable selections $\widehat{\mu}_{a,s_j} \in \arg\min_{q\in\mathcal{M}} n_a^{-1}\sum_{i=1}^{n_a} d_{\mathcal{M}}(O_{i,a,s_j},q)^2$ for $a \in \{T,C\}$ and $j \in \{0,1\}$. Let $\mathbf{m}_{\mathrm{G}} := (\mu_{T,s_0},\mu_{T,s_1},\mu_{C,s_0},\mu_{C,s_1})$ and $\widehat{\mathbf{m}}_{\mathrm{G}} := (\widehat{\mu}_{T,s_0},\widehat{\mu}_{T,s_1},\widehat{\mu}_{C,s_0},\widehat{\mu}_{C,s_1})$. Define the product chart $\Phi_{\mathrm{G}}(m_{T,s_0},m_{T,s_1},m_{C,s_0},m_{C,s_1}) := (\Psi_{\mathrm{PG}}(m_{T,s_0}),\Psi_{\mathrm{PG}}(m_{T,s_1}),\Psi_{\mathrm{PG}}(m_{C,s_0}),\Psi_{\mathrm{PG}}(m_{C,s_1})) \in \mathbb{R}^{4d}$. The lemma states the limit of the four sample Fréchet means in this product chart.

Lemma 4.1 (Joint CLT for pre-period Fréchet means)**.** *Under Assumptions 2.1, 4.1 and 4.2, let* $\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})$ *equal a fixed deterministic vector of* $\mathbb{R}^{4d}$ *on the event that some* $\widehat{\mu}_{a,s_j}$ *lies outside* $\mathcal{U}_{\mathrm{PG}}$*. Then there exists a covariance matrix* $\Xi_{\mathrm{G}} \in \mathbb{R}^{4d\times 4d}$ *such that*

$$\sqrt{n}\left[\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) - \Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right] \Rightarrow \mathcal{N}(0,\Xi_{\mathrm{G}}).$$

In Lemma 4.1, the dependence between the two pre-period outcomes of a unit enters Ξ_{G} through the blocks that connect the two periods of a group. Because the treated and control samples are independent, Ξ_{G} is block diagonal across the two groups, and the inverses ρ_T^{-1} and ρ_C^{-1} of the limiting group shares in Assumption 4.1 scale its two blocks. The proof is given in Appendix B.1.

For $x_0,x_1,z_0,z_1 \in \mathcal{U}_{\mathrm{PG}}$, define the smooth GDiD pretrend contrast map

$$g_{\mathrm{G}}(x_0,x_1,z_0,z_1) := B_{\mathrm{ref}}^{-1}\Pi_{\eta_{x_0\to\mu_{\mathrm{ref}}}}\left[\mathrm{Log}_{x_0}(x_1) - \Pi_{\eta_{z_0\to x_0}}\left\{\mathrm{Log}_{z_0}(z_1)\right\}\right].$$

The population GDiD pretrend discrepancy is $\delta_0^{\mathrm{G}} := g_{\mathrm{G}}(\mu_{T,s_0},\mu_{T,s_1},\mu_{C,s_0},\mu_{C,s_1}) \in \mathbb{R}^d$. The GDiD pretrend null hypothesis is $H_0^{\mathrm{GDiD}} : \delta_0^{\mathrm{G}} = 0$. Let $\mathcal{E}_n^{\mathrm{G}} := \{\widehat{\mu}_{a,s_j} \in \mathcal{U}_{\mathrm{PG}} \text{ for all } a \in \{T,C\}, j \in \{0,1\}\}$. On $\mathcal{E}_n^{\mathrm{G}}$, define the local sample GDiD pretrend discrepancy by $\widehat{\delta}_{n,\mathrm{loc}}^{\mathrm{G}} := g_{\mathrm{G}}(\widehat{\mu}_{T,s_0},\widehat{\mu}_{T,s_1},\widehat{\mu}_{C,s_0},\widehat{\mu}_{C,s_1})$. The map g_{G} need not be defined on $(\mathcal{E}_n^{\mathrm{G}})^c$, because the logarithmic maps and parallel transports entering its definition may fail to be uniquely defined outside the common strongly convex neighborhood. To obtain an $\mathbb{R}^d$-valued random variable on the whole sample space for the

purpose of stating the asymptotic result, fix an arbitrary deterministic vector $r_{\mathrm{G}} \in \mathbb{R}^d$ and define the completed estimator by $\widehat{\delta}_n^{\mathrm{G}} := \widehat{\delta}_{n,\mathrm{loc}}^{\mathrm{G}}$ on $\mathcal{E}_n^{\mathrm{G}}$ and $\widehat{\delta}_n^{\mathrm{G}} := r_{\mathrm{G}}$ on $(\mathcal{E}_n^{\mathrm{G}})^c$. With these definitions, the completed discrepancy satisfies the following central limit theorem, which also gives the limit of a Wald statistic.

Theorem 4.1 (Pretrend GDiD)**.** *Under Assumptions 2.1, 4.1, and 4.2, there exists a covariance matrix $\Sigma_{\mathrm{G}} \in \mathbb{R}^{d\times d}$ such that $\sqrt{n}\left(\widehat{\delta}_n^{\mathrm{G}} - \delta_0^{\mathrm{G}}\right) \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{G}})$. Equivalently, if $D_{\mathrm{G}} := D\left[g_{\mathrm{G}} \circ \Phi_{\mathrm{G}}^{-1}\right]\{\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\}$, then $\Sigma_{\mathrm{G}} = D_{\mathrm{G}}\Xi_{\mathrm{G}}D_{\mathrm{G}}^{\top}$. In particular, under H_0^{GDiD}, $\sqrt{n}\,\widehat{\delta}_n^{\mathrm{G}} \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{G}})$. If Σ_{G} is positive definite and an estimator $\widehat{\Sigma}_{\mathrm{G}}$ satisfies $\widehat{\Sigma}_{\mathrm{G}} \to_p \Sigma_{\mathrm{G}}$, then the Wald statistic $W_n^{\mathrm{G}} := n(\widehat{\delta}_n^{\mathrm{G}})^{\top}\widehat{\Sigma}_{\mathrm{G}}^{-1}\widehat{\delta}_n^{\mathrm{G}}$ satisfies, under H_0^{GDiD}, $W_n^{\mathrm{G}} \Rightarrow \chi_d^2$.*

Theorem 4.1 centers the completed discrepancy at the population discrepancy δ_0^{G}, which compares the treated group's pre-period mean movement with the control group's movement after the latter is parallel transported to the treated baseline mean, and gives its normal limit with covariance Σ_{G}. To test H_0^{GDiD}, one computes W_n^{G} with a consistent estimator $\widehat{\Sigma}_{\mathrm{G}}$ of a positive definite Σ_{G} and rejects at a given nominal level when W_n^{G} exceeds the corresponding upper quantile of the χ_d^2 distribution. The proof is given in Appendix B.2.

4.2. **TDiD pretrend test.** The TDiD pretrend test is based on unit-level increments in the fixed common tangent-space chart. It tests whether the mean increment $\Psi_{\mathrm{ref}}(O_{i,a,s_1}) - \Psi_{\mathrm{ref}}(O_{i,a,s_0})$ is the same for the treated and control groups during the pre-treatment periods. The TDiD test uses the common chart Ψ_{ref} of Section 2.2 and the following support condition.

Assumption 4.3 (Pretrend common-chart support)**.** *There exist an open normal neighborhood $\mathcal{U}_{\mathrm{PT}} \subset \mathcal{U}_{\mathrm{ref}}$ of μ_{ref} and a compact set $\mathcal{K}_{\mathrm{PT}} \subset \mathcal{U}_{\mathrm{PT}}$ such that $\mathbb{P}\left(O_{i,a,s_j} \in \mathcal{K}_{\mathrm{PT}}\right) = 1$ for $a \in \{T, C\}$ and $j \in \{0, 1\}$.*

Assumption 4.3 keeps every pre-period outcome in a compact subset of a normal neighborhood of μ_{ref} inside the chart domain. Unlike Assumption 4.2, it involves neither Fréchet means nor curvature, because the TDiD test works with ordinary means of chart coordinates.

Under Assumption 4.3, define the common-chart pre-period variables by $Y_{i,a,s_j} := \Psi_{\mathrm{ref}}(O_{i,a,s_j}) = B_{\mathrm{ref}}^{-1}\operatorname{Log}_{\mu_{\mathrm{ref}}}(O_{i,a,s_j}) \in \mathbb{R}^d$ for $a \in \{T, C\}$ and $j \in \{0, 1\}$. Let $Y_{a,s_j} := \Psi_{\mathrm{ref}}(O_{a,s_j}) = B_{\mathrm{ref}}^{-1}\operatorname{Log}_{\mu_{\mathrm{ref}}}(O_{a,s_j})$ denote the corresponding generic random vector. Since $\mathcal{K}_{\mathrm{PT}}$ is compact and Ψ_{ref} is continuous on $\mathcal{U}_{\mathrm{PT}}$, these random vectors are bounded. Define the unit-level pre-period increments by $\Delta_{i,a} := Y_{i,a,s_1} - Y_{i,a,s_0}$ for $a \in \{T, C\}$, and the corresponding generic increments

by $\Delta_a := Y_{a,s_1} - Y_{a,s_0}$. The group-specific sample mean increments are $\overline{\Delta}_a := \frac{1}{n_a}\sum_{i=1}^{n_a}\Delta_{i,a}$ for $a \in \{T, C\}$. The population TDiD pretrend discrepancy is $\delta_0^{\mathrm{T}} := \mathbb{E}[\Delta_T] - \mathbb{E}[\Delta_C] \in \mathbb{R}^d$. The TDiD pretrend null hypothesis is $H_0^{\mathrm{TDiD}} : \delta_0^{\mathrm{T}} = 0$. The sample TDiD pretrend discrepancy is $\widehat{\delta}_n^{\mathrm{T}} := \overline{\Delta}_T - \overline{\Delta}_C$. The sample discrepancy then satisfies the following central limit theorem.

Theorem 4.2 (Pretrend TDiD)**.** *Under Assumptions 2.1, 4.1, and 4.3, let $V_a := \mathrm{Var}(\Delta_a)$ for $a \in \{T, C\}$. Then $\sqrt{n}\left(\widehat{\delta}_n^{\mathrm{T}} - \delta_0^{\mathrm{T}}\right) \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{T}})$, where $\Sigma_{\mathrm{T}} = \rho_T^{-1}V_T + \rho_C^{-1}V_C$. In particular, under H_0^{TDiD}, $\sqrt{n}\,\widehat{\delta}_n^{\mathrm{T}} \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{T}})$.*

Theorem 4.2 yields a Wald test once Σ_{T} is estimated consistently. Define the plug-in covariance estimator by $\widehat{V}_a := \frac{1}{n_a}\sum_{i=1}^{n_a}(\Delta_{i,a} - \overline{\Delta}_a)(\Delta_{i,a} - \overline{\Delta}_a)^\top$ for $a \in \{T, C\}$, and $\widehat{\Sigma}_{\mathrm{T}} := \frac{n}{n_T}\widehat{V}_T + \frac{n}{n_C}\widehat{V}_C$. If, in addition, Σ_{T} is positive definite, then the Wald statistic $W_n^{\mathrm{T}} := n(\widehat{\delta}_n^{\mathrm{T}})^\top\widehat{\Sigma}_{\mathrm{T}}^{-1}\widehat{\delta}_n^{\mathrm{T}}$ satisfies, under H_0^{TDiD}, $W_n^{\mathrm{T}} \Rightarrow \chi_d^2$. The test rejects H_0^{TDiD} at a given nominal level when W_n^{T} exceeds the corresponding upper quantile of the χ_d^2 distribution. The proof is given in Appendix B.3.

5. Identification and estimation of MCiC

This section presents the identification and estimation of manifold changes-in-changes (MCiC). Section 5.1 identifies the untreated counterfactual distribution of the treated group, Section 5.2 defines an estimator of this distribution, and Section 5.3 derives the convergence rate of the estimator.

5.1. Identification.

This subsection identifies the untreated counterfactual distribution of the treated group in the post-treatment period. Throughout this subsection, λ^d denotes the d-dimensional Lebesgue measure on $\mathbb{R}^d$, $\ll$ denotes absolute continuity, and $\mathcal{P}_2(\mathbb{R}^d)$ denotes the set of Borel probability measures on $\mathbb{R}^d$ with finite second moments. The Euclidean inner product and norm are denoted by $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$. The following assumption states the primitive conditions through a latent variable for each group and two structural maps, one for each period.

Assumption 5.1 (Primitive conditions for MCiC)**.** *There exist an open normal neighborhood $\mathcal{U}_{\mathrm{MCiC}} \subset \mathcal{U}_{\mathrm{ref}}$ of μ_{ref} and a compact set $\mathcal{K}_{\mathrm{MCiC}} \subset \mathcal{U}_{\mathrm{MCiC}}$. Define $\mathcal{Y}_{\mathrm{ref}} := \Psi_{\mathrm{ref}}(\mathcal{K}_{\mathrm{MCiC}}) \subset \mathbb{R}^d$. There exist compact sets $\mathcal{Y}_0, \mathcal{Y}_1 \subset \mathcal{Y}_{\mathrm{ref}}$ where $\mathcal{Y}_0$ is convex. There exist a compact set $\mathcal{Z} \subset \mathbb{R}^d$, random vectors Z_C and Z_T taking values in $\mathcal{Z}$, and measurable maps $\widetilde{h}_0, \widetilde{h}_1 : \mathcal{Z} \to \mathcal{K}_{\mathrm{MCiC}}$. Let Q_C and Q_T denote the laws of Z_C and Z_T, respectively. For $j = 0, 1$, define the chart-mapped structural functions $h_j := \Psi_{\mathrm{ref}} \circ \widetilde{h}_j$, such that $h_0 : \mathcal{Z} \to \mathcal{Y}_0$ and $h_1 : \mathcal{Z} \to \mathcal{Y}_1$. The following conditions hold.*

(1) *The untreated structural maps are common across groups:* $O_{C,0} = \widetilde{h}_0(Z_C)$, $O_{C,1} = \widetilde{h}_1(Z_C)$, $O_{T,0} = \widetilde{h}_0(Z_T)$, *and* $O_{T,1}(0) = \widetilde{h}_1(Z_T)$ *almost surely.*

(2) *The map* $h_0 : \mathcal{Z} \to \mathcal{Y}_0$ *is bijective, and its inverse* $h_0^{-1} : \mathcal{Y}_0 \to \mathcal{Z}$ *is measurable.*

(3) *The pair* (h_0, h_1) *is cyclically monotone on* $\mathcal{Z}$*: for every positive integer* r *and every cycle* $z_1, \ldots, z_r, z_{r+1} \equiv z_1 \in \mathcal{Z}$, $\sum_{j=1}^{r} \langle h_0(z_j), h_1(z_j) - h_1(z_{j+1}) \rangle \geq 0$.

(4) *The control pre-treatment chart law induced by the latent model is absolutely continuous:* $h_{0\#}Q_C \ll \lambda^d$.

(5) *The latent overlap condition* $Q_T \ll Q_C$ *holds.*

Assumption 5.1 should be read as a local, chart-level version of a nonlinear untreated time-evolution model. The latent variable Z_a indexes units in group a, the map h_0 gives the baseline outcome in the common tangent-space chart, and the map h_1 gives the untreated post-period outcome in the same chart. Conditions (1)–(3) describe the untreated time evolution through these structural maps, and conditions (4) and (5) restrict the control baseline law and the treated latent law. Condition (1) states that treated and control units with the same latent value have the same untreated outcomes in both periods, so the two groups share the common untreated transition map once their baseline positions are represented in the chart. Condition (2) states that each baseline position in $\mathcal{Y}_0$ corresponds to exactly one latent value, so this map is defined on $\mathcal{Y}_0$. Condition (3) restricts how the untreated evolution can reorder units. Condition (4) requires the control baseline law in the chart to have a density, which makes the optimal transport from the control baseline law to the control post-period law unique. Condition (5) requires every set of latent values with probability zero in the control group to have probability zero in the treated group as well, so the map identified from the control group also applies to the treated baseline law.

In particular, condition (3) is the multivariate analogue of a monotone untreated evolution. In one dimension it says that the untreated evolution preserves the ranks of units, as in the changes-in-changes model of Athey and Imbens (2006). In several dimensions it says that no reassignment of post-period positions within a finite cycle of units lowers their total squared displacement, which is the property that characterizes optimal transport for the squared Euclidean cost; see, for example, Villani (2003). Definition 1 of Torous et al. (2024) calls this condition cyclical comonotonicity, and that paper imposes it on the production functions of the untreated outcomes in its latent model, which extends the changes-in-changes model to multivariate outcomes. Condition (3) imposes it on

the chart-mapped structural functions h_0 and h_1. It is an identifying restriction on the untreated temporal evolution and does not follow from the two observed marginal laws.

The next proposition collects the implications of Assumption 5.1 that the identification argument uses. To state it, we write the common tangent-space pushforwards of the manifold-valued laws as $\nu_{C,0} := (\Psi_{\mathrm{ref}})_{\#}P_{C,0}$, $\nu_{C,1} := (\Psi_{\mathrm{ref}})_{\#}P_{C,1}$, $\nu_{T,0} := (\Psi_{\mathrm{ref}})_{\#}P_{T,0}$, and $\nu^{(0)}_{T,1} := (\Psi_{\mathrm{ref}})_{\#}P^{(0)}_{T,1}$, and their supports as $\mathcal{S}_{C,0} := \mathrm{supp}(\nu_{C,0})$, $\mathcal{S}_{C,1} := \mathrm{supp}(\nu_{C,1})$, $\mathcal{S}_{T,0} := \mathrm{supp}(\nu_{T,0})$, and $\mathcal{S}^{(0)}_{T,1} := \mathrm{supp}(\nu^{(0)}_{T,1})$. Since $\mathcal{Y}_{\mathrm{ref}}$ is compact, the measures $\nu_{C,0}$, $\nu_{C,1}$, $\nu_{T,0}$, and $\nu^{(0)}_{T,1}$ belong to $\mathcal{P}_2(\mathbb{R}^d)$. Under Assumption 5.1, we also define the common untreated transition map from period 0 to period 1 on the common tangent-space chart by $T^{(0)}_{0\to1} := h_1 \circ h_0^{-1} : \mathcal{Y}_0 \to \mathcal{Y}_1$.

Proposition 5.1 (Implications of the primitive conditions)**.** *Under Assumptions 2.1 and 5.1, the map $T^{(0)}_{0\to1} : \mathcal{Y}_0 \to \mathcal{Y}_1$ is well-defined and measurable. Moreover, the following statements hold.*

(1) $\nu_{C,1} = (T^{(0)}_{0\to1})_{\#}\nu_{C,0}$.

(2) $\nu_{C,0} \ll \lambda^d$.

(3) *The map $T^{(0)}_{0\to1}$ is cyclically monotone: for every positive integer r and every cycle $y_1, \ldots, y_r, y_{r+1} \equiv y_1 \in \mathcal{Y}_0$, $\sum_{j=1}^{r} \left\langle y_j, T^{(0)}_{0\to1}(y_j) - T^{(0)}_{0\to1}(y_{j+1}) \right\rangle \geq 0$.*

(4) $\nu_{T,0} \ll \nu_{C,0}$ *and* $\mathcal{S}_{T,0} \subset \mathcal{S}_{C,0} \subset \mathcal{Y}_0$.

(5) $\nu^{(0)}_{T,1} = (T^{(0)}_{0\to1})_{\#}\nu_{T,0}$.

Proposition 5.1 provides the link between the primitive structural assumptions and the Brenier map used in the MCiC identification formula. The link involves two maps on the common tangent-space chart that move baseline positions to post-period positions, and they are defined from different objects. The common untreated transition map $T^{(0)}_{0\to1} = h_1 \circ h_0^{-1}$ is defined through the structural maps of Assumption 5.1, which act on unobserved latent variables, so it is not directly observed. The Brenier map $B^{C}_{0\to1}$ is defined from the two observed control chart laws alone, as the map that pushes $\nu_{C,0}$ forward to $\nu_{C,1}$ with the smallest expected squared Euclidean displacement.

Statements (1), (2), and (3) imply that $T^{(0)}_{0\to1}$ is the Brenier transport from $\nu_{C,0}$ to $\nu_{C,1}$ for the squared Euclidean cost. Indeed, by (1), $T^{(0)}_{0\to1}$ pushes the control pre-treatment law to the control post-treatment law: $\nu_{C,1} = (T^{(0)}_{0\to1})_{\#}\nu_{C,0}$. By (3), $T^{(0)}_{0\to1}$ is cyclically monotone, so the coupling $(\mathrm{id}, T^{(0)}_{0\to1})_{\#}\nu_{C,0}$ is optimal for the squared Euclidean cost. By (2), the source law is absolutely continuous, $\nu_{C,0} \ll \lambda^d$, and hence Brenier's theorem yields a $\nu_{C,0}$-almost everywhere unique optimal transport map from $\nu_{C,0}$ to $\nu_{C,1}$. Therefore, the Brenier map $B^{C}_{0\to1}$ identified from the observed control laws satisfies $B^{C}_{0\to1} = T^{(0)}_{0\to1}$ $\nu_{C,0}$-almost everywhere.

Statements (4) and (5) then allow this control-identified map to be applied to the treated pre-treatment law. By (4), $\nu_{T,0} \ll \nu_{C,0}$, so the equality $B^C_{0\to1} = T^{(0)}_{0\to1}$ also holds $\nu_{T,0}$-almost everywhere. Combining this with (5) gives $\nu^{(0)}_{T,1} = (T^{(0)}_{0\to1})_{\#}\nu_{T,0} = (B^C_{0\to1})_{\#}\nu_{T,0}$. Thus, the proposition shows both why the control-group Brenier map is identified and why applying it to the treated pre-treatment distribution recovers the untreated counterfactual distribution. The proof of Proposition 5.1 is given in Appendix C.1. Combining Proposition 5.1 with Brenier's theorem gives the identification result.

Theorem 5.1 (MCiC identification on $\mathcal{M}$)**.** *Under Assumptions 2.1 and 5.1, the observed distribution identifies $P_{C,0}$, $P_{C,1}$, and $P_{T,0}$, and hence also $\nu_{C,0}$, $\nu_{C,1}$, and $\nu_{T,0}$. Let $B^C_{0\to1} : \mathbb{R}^d \to \mathbb{R}^d$ denote the Brenier map transporting $\nu_{C,0}$ to $\nu_{C,1}$ for the squared Euclidean cost. Then $B^C_{0\to1}$ exists, is unique $\nu_{C,0}$-almost everywhere, is identified from $\nu_{C,0}$ and $\nu_{C,1}$, and satisfies $B^C_{0\to1} = T^{(0)}_{0\to1}$ both $\nu_{C,0}$- and consequently $\nu_{T,0}$-almost everywhere. The untreated counterfactual distribution on the common tangent-space chart is therefore identified as $\nu^{(0)}_{T,1} = (B^C_{0\to1})_{\#}\nu_{T,0} = (T^{(0)}_{0\to1})_{\#}\nu_{T,0}$. The corresponding untreated counterfactual distribution on the manifold is identified as $P^{(0)}_{T,1} = (\Psi^{-1}_{\mathrm{ref}})_{\#}\big((B^C_{0\to1})_{\#}\nu_{T,0}\big) = (\Psi^{-1}_{\mathrm{ref}})_{\#}\big((T^{(0)}_{0\to1})_{\#}\nu_{T,0}\big)$.*

Theorem 5.1 expresses the untreated counterfactual distribution through the three observed chart laws. The Brenier map determined by $\nu_{C,0}$ and $\nu_{C,1}$ is applied to $\nu_{T,0}$, and the inverse chart carries the resulting chart law back to $\mathcal{M}$. Figure 9 summarizes the maps between the laws in Theorem 5.1. Because the whole counterfactual distribution is identified, so is every functional of it, such as its Fréchet mean or its chart mean. The proof is given in Appendix C.2.

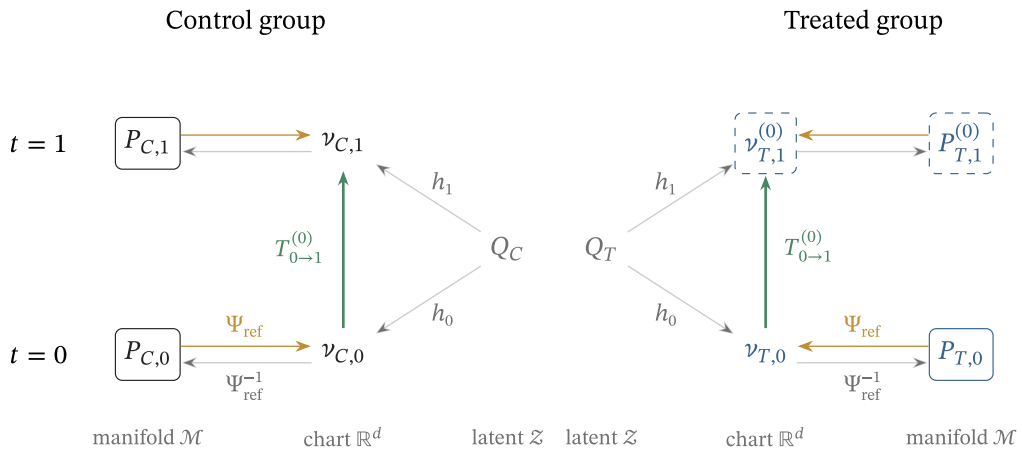


FIGURE 9. Maps between laws in the MCiC identification of Theorem 5.1. Laws in solid boxes are observed, and laws in dashed boxes are counterfactual. Orange arrows are the chart Ψ_{ref}, and gray horizontal arrows are its inverse. The structural maps h_0 and h_1 of Assumption 5.1 push the latent laws Q_C and Q_T to the chart laws. The green arrows are the common untreated transition map $T^{(0)}_{0\to1} = h_1 \circ h_0^{-1}$, which equals the control Brenier map $B^C_{0\to1}$.

5.2. **Estimation.** This subsection estimates the counterfactual distribution identified in Theorem 5.1. It defines the empirical transport estimator and its one-nearest-neighbor extension. By Theorem 5.1, the tangent-space untreated counterfactual distribution is $\nu^{(0)}_{T,1} = (B^C_{0\to1})_{\#}\nu_{T,0}$, and the corresponding manifold-valued distribution is $P^{(0)}_{T,1} = (\Psi^{-1}_{\text{ref}})_{\#}\nu^{(0)}_{T,1} = (\Psi^{-1}_{\text{ref}})_{\#}\big((B^C_{0\to1})_{\#}\nu_{T,0}\big)$. The estimator replaces $B^C_{0\to1}$ by an empirical transport map and $\nu_{T,0}$ by the empirical law of the treated baseline sample.

Let $Y_{C,0,1}, \dots, Y_{C,0,n}$, $Y_{C,1,1}, \dots, Y_{C,1,\ell}$, and $Y_{T,0,1}, \dots, Y_{T,0,m}$ denote the observed samples on the common tangent-space chart. Define the empirical distributions $\widehat{\nu}_{C,0,n} := \frac{1}{n}\sum_{i=1}^{n} \delta_{Y_{C,0,i}}$, $\widehat{\nu}_{C,1,\ell} := \frac{1}{\ell}\sum_{j=1}^{\ell} \delta_{Y_{C,1,j}}$, and $\widehat{\nu}_{T,0,m} := \frac{1}{m}\sum_{k=1}^{m} \delta_{Y_{T,0,k}}$. For probability measures α and β, let $\Pi(\alpha, \beta)$ denote the set of couplings with marginals α and β. To estimate the temporal transport in the control group, solve the empirical OT problem $\widehat{\pi}_{\text{time}} \in \arg\min_{\pi \in \Pi(\widehat{\nu}_{C,0,n}, \widehat{\nu}_{C,1,\ell})} \int \|y - y'\|^2 \, \pi(\mathrm{d}y, \mathrm{d}y')$. If the empirical OT problem has multiple optimal plans, one is chosen by a fixed deterministic tie-breaking rule. Writing the selected optimal plan as a mass matrix $(\widehat{\pi}_{ij})$, its barycentric projection on the

empirical source support is

$$\widehat{B}^{C}_{0\to1,n,\ell}(Y_{C,0,i}) := \frac{\sum_{j=1}^{\ell} \widehat{\pi}_{ij} Y_{C,1,j}}{\sum_{j=1}^{\ell} \widehat{\pi}_{ij}}, \qquad i = 1, \dots, n.$$

The empirical barycentric projection is defined only at the control source observations. We use its one-nearest-neighbor (1NN) extension and call the resulting counterfactual estimator 1NN OT. For $y \in \mathcal{Y}_0$, let $i_n(y)$ be the smallest index minimizing $\|y - Y_{C,0,i}\|$ over $i = 1, \dots, n$, and set

$$\widehat{B}^{C,\mathrm{1NN}}_{0\to1,n,\ell}(y) := \widehat{B}^{C}_{0\to1,n,\ell}\{Y_{C,0,i_n(y)}\}.$$

In words, a chart point y is first matched to its nearest control baseline observation $Y_{C,0,i_n(y)}$, and it then receives the barycentric image of that observation. The extension is therefore constant on each nearest-neighbor cell of the control baseline sample. Figure 10 illustrates the rule.

The smallest-index rule makes the extension measurable. Under the density condition of Assumption 5.2 below, the source observations are distinct almost surely, so the extension agrees with the barycentric projection at each of them. Each barycentric image is an average of control post-period observations, so it lies in the output support whenever this support is convex. Applying the extension to the treated baseline empirical law gives the 1NN OT estimators of the counterfactual distributions in the chart and on the manifold,

$$\widehat{\nu}^{(0),\mathrm{1NN}}_{T,1,m} := (\widehat{B}^{C,\mathrm{1NN}}_{0\to1,n,\ell})_{\#}\widehat{\nu}_{T,0,m}, \qquad \widehat{P}^{(0),\mathrm{1NN}}_{T,1,m} := (\Psi_{\mathrm{ref}}^{-1})_{\#}\widehat{\nu}^{(0),\mathrm{1NN}}_{T,1,m}.$$

Thus the 1NN OT (Time) estimate reported in Sections 6 and 7 uses the empirical control time plan, its barycentric images, and the nearest control baseline neighbor of each treated baseline observation. The 1NN OT (State) comparison applies the same algorithm to a baseline cross-group plan. The 1NN extension is piecewise constant and generally discontinuous, and Theorem 5.2 assumes no Lipschitz property of this estimated extension.

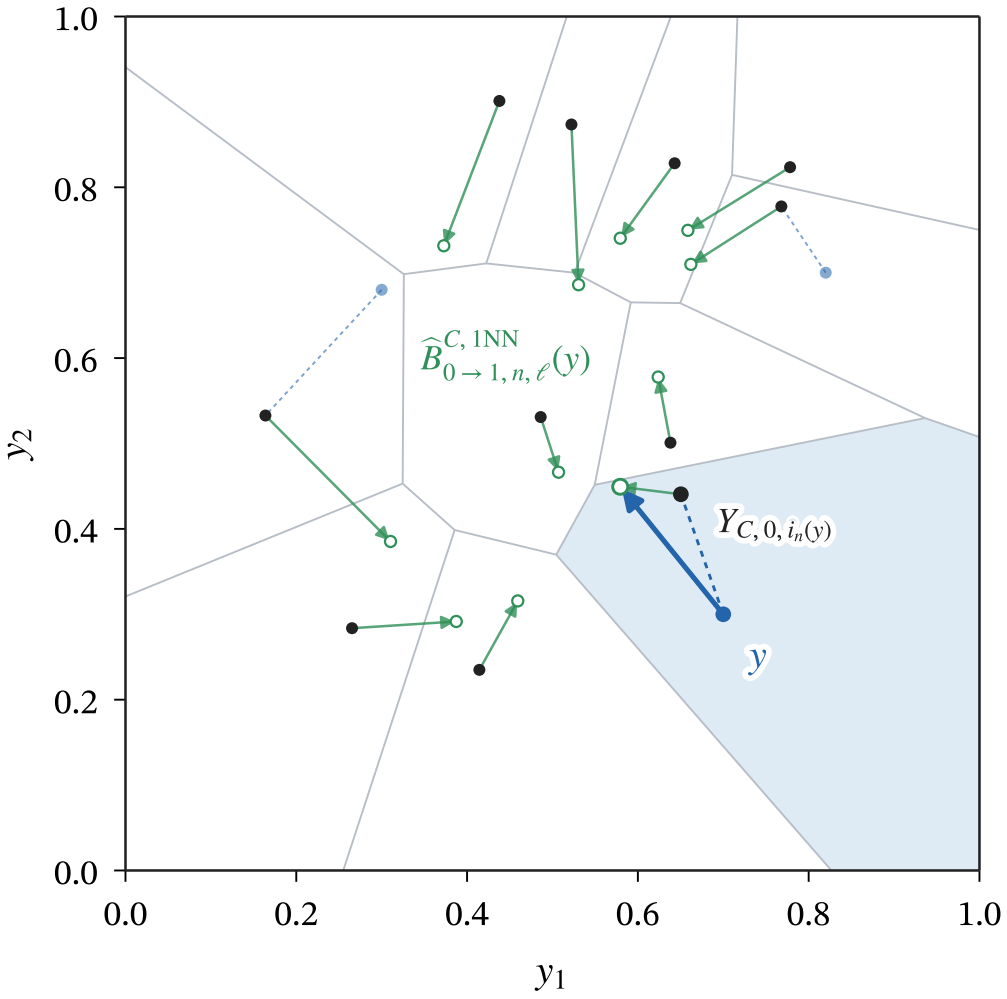


FIGURE 10. The one-nearest-neighbor extension of the empirical barycentric projection in a two-dimensional chart. Black points are control baseline observations $Y_{C,0,i}$, and each green arrow ends at its barycentric image (hollow green point) under an exact empirical OT plan with equal sample sizes, so that the plan is an assignment. Gray lines bound the nearest-neighbor cells of the control baseline sample. The treated baseline point y (blue) lies in the shaded cell of $Y_{C,0,i_n(y)}$, to which the dotted line connects it, and the blue arrow shows that $\widehat{B}^{C,1\mathrm{NN}}_{0\to 1,n,\ell}(y)$ is the barycentric image of $Y_{C,0,i_n(y)}$. Faint blue points are other treated baseline observations. The samples are synthetic.

5.3. **Convergence rate.** This subsection derives the convergence rate of the 1NN OT estimator in the 2-Wasserstein distance. Let W_2 denote the 2-Wasserstein distance on $\mathbb{R}^d$ induced by the Euclidean norm, and let $W_{2,\mathcal{M}}$ denote the 2-Wasserstein distance on $\mathcal{M}$ induced by $d_{\mathcal{M}}$. The proof of the rate in Theorem 5.2 below reduces the manifold Wasserstein error to a Euclidean error in the chart. This reduction uses the following Lipschitz property of the inverse chart.

Lemma 5.1 (Lipschitz continuity of the inverse chart)**.** *Suppose that Assumption 2.1 holds, and let $\mathcal{Y} \subset \mathbb{R}^d$ be compact and convex. Then $C_\Psi := \sup_{w\in\mathcal{Y}} \|D \operatorname{Exp}_{\mu_{\mathrm{ref}}}|_{B_{\mathrm{ref}}w} \circ B_{\mathrm{ref}}\|_{\mathrm{op}}$ is finite, and $d_{\mathcal{M}}\{\operatorname{Exp}_{\mu_{\mathrm{ref}}}(B_{\mathrm{ref}}u), \operatorname{Exp}_{\mu_{\mathrm{ref}}}(B_{\mathrm{ref}}v)\} \le C_\Psi\|u-v\|$ for all $u, v \in \mathcal{Y}$. In particular, Ψ_{ref}^{-1} is C_Ψ-Lipschitz on every compact convex subset of the chart image.*

The proof of Lemma 5.1 is given in Appendix C.3. In addition to Assumptions 2.1 and 5.1, Theorem 5.2 uses conditions on the sampling, on the supports and densities of the chart laws, on the Brenier potential, and on tie-breaking, which the following assumption collects.

Assumption 5.2 (Conditions for the one-nearest-neighbor OT rate)**.** *The following conditions hold.*

(1) *Each of the samples $Y_{C,0,1}, \ldots, Y_{C,0,n}$, $Y_{C,1,1}, \ldots, Y_{C,1,\ell}$, and $Y_{T,0,1}, \ldots, Y_{T,0,m}$ is i.i.d., and the three samples are mutually independent.*

(2) *For some $a \in \mathbb{R}^d$ and $s > 0$, the supports of $\nu_{C,0}$, $\nu_{C,1}$, and $\nu_{T,0}$ are the same cube $a+s[0,1]^d$. Conditions (3) and (4) are stated in the normalized coordinates $A(y) := (y-a)/s$, in which this cube is $[0,1]^d$.*

(3) *Each of the three laws has a Lebesgue density $g_{C,0}$, $g_{C,1}$, or $g_{T,0}$ satisfying $0 < c \le g_{a,t}(y) \le C < \infty$ almost everywhere on the cube, for common fixed constants c, C.*

(4) *The Brenier map $B_{0\to1}^C = \nabla\phi$ has a C^2 potential on the cube and, for a fixed $L \ge 1$, satisfies $L^{-1}I_d \preceq \nabla^2\phi(y) \preceq LI_d$ there.*

(5) *The empirical OT plan and the 1NN index are chosen with fixed measurable tie rules.*

Below, we use the normalized coordinates of condition (2). Because the same A acts on all samples, 1NN indices and empirical barycentric images transform equivariantly, so the resulting manifold estimator is unchanged. The inverse chart in these coordinates is the original $\Psi_{\mathrm{ref}}^{-1} \circ A^{-1}$, and for brevity we denote this composite again by Ψ_{ref}^{-1}. By condition (2) and Assumption 5.1, $A^{-1}([0,1]^d)$ is the common support of the three chart laws in the original coordinates and lies in $\mathcal{Y}_{\mathrm{ref}}$, so it is a compact convex subset of the chart image. Applying Lemma 5.1 to this set and using

$A^{-1}(y) = a + sy$ shows that the normalized inverse chart is Lipschitz on the cube with constant s times the constant of Lemma 5.1. In Theorem 5.2 and Appendix C.4, C_Ψ always denotes this normalized constant. Condition (3) implies $\|f_{T|C}\|_\infty \le C/c$, where $f_{T|C} = d\nu_{T,0}/d\nu_{C,0}$.

Together, these conditions are sufficient for the sharper cube branch of Proposition 15 of Manole et al. (2024) and for the empirical Wasserstein bound used in the proof of Theorem 5.2. Under these conditions, the 1NN OT estimator satisfies the following bound.

Theorem 5.2 (General-dimensional rate for the 1NN OT estimator)**.** *Under Assumptions 2.1, 5.1, and 5.2, for constants $K_1, K_2 < \infty$ depending only on d, L, c, and C,*

$$\mathbb{E}W_{2,\mathcal{M}}\Big(\widehat{P}^{(0),1\mathrm{NN}}_{T,1,m}, P^{(0)}_{T,1}\Big) \le C_\Psi \Bigg[K_1 \log(en) \begin{cases} (n \wedge \ell)^{-1/2}, & d = 1, \\ (n \wedge \ell)^{-1/2} \log\{e(n \wedge \ell)\}, & d = 2, \\ (n \wedge \ell)^{-1/d}, & d \ge 3, \end{cases} + K_2 L \begin{cases} m^{-1/2}, & d = 1, \\ m^{-1/2} \log(em), & d = 2, \\ m^{-1/d}, & d \ge 3. \end{cases} \Bigg].$$

The first term is the source-sample 1NN transport-map estimation error, and the second is the treated-baseline empirical-distribution error. If $n \asymp \ell \asymp m \asymp N$, the bound is $O(N^{-1/2} \log N)$ for $d = 1$, $O\{N^{-1/2}(\log N)^2\}$ for $d = 2$, and $O(N^{-1/d} \log N)$ for $d \ge 3$. For unequal sample sizes, the displayed expression is an upper bound, and it implies convergence along any sequence on which both displayed terms tend to zero. The proof is in Appendix C.4.

6. Simulation

We use a stylized voting design with three vote-share categories. The purpose is to isolate a common nonlinear temporal transition on the simplex and to examine how different DID-type estimators behave when baseline treated and control distributions are compositionally imbalanced. Section 6.1 represents compositional outcomes on the sphere and describes the compositional DID benchmark, Section 6.2 describes the data-generating process, and Section 6.3 reports the results.

6.1. **Compositional data as a manifold.**

This subsection specializes the preceding framework to compositional outcomes. Let $K \ge 2$ be the number of categories, and define the strictly positive

simplex by

$$\Delta_{++}^{K-1} := \left\{ x = (x_1, \ldots, x_K)^\top \in \mathbb{R}_{>0}^K : \sum_{k=1}^K x_k = 1 \right\}.$$

For unit i and period $t \in \{0,1\}$, let $X_{i,t}(0),\ X_{i,t}(1) \in \Delta_{++}^{K-1}$ denote the potential compositional outcomes. The observed composition is

$$X_{i,t} = \begin{cases} X_{i,t}(0), & G_i = C \text{ or } t = 0, \\ X_{i,t}(1), & G_i = T \text{ and } t = 1. \end{cases}$$

The strict positivity condition is used to keep the square-root embedding and the componentwise ratios below away from the boundary. We represent compositions on the unit sphere through the square-root embedding

$$\varphi_{\text{comp}} : \Delta_{++}^{K-1} \to \mathbb{S}_{++}^{K-1}, \qquad \varphi_{\text{comp}}(x) = x^{1/2} := (\sqrt{x_1}, \ldots, \sqrt{x_K})^\top,$$

where $\mathbb{S}^{K-1} := \{o \in \mathbb{R}^K : \|o\|_2 = 1\}, \mathbb{S}_{++}^{K-1} := \mathbb{S}^{K-1} \cap \mathbb{R}_{>0}^K$. In this example, the ambient manifold is $\mathcal{M}_{\text{comp}} := \mathbb{S}^{K-1}$, equipped with the standard spherical geodesic distance $d_{\mathcal{M}_{\text{comp}}}(o, o') = \arccos(o^\top o')$. The manifold-valued outcome in the notation of the preceding sections is $O_{i,t}(z) = \varphi_{\text{comp}}\{X_{i,t}(z)\}, O_{i,t} = \varphi_{\text{comp}}(X_{i,t})$. For $o \in \mathbb{S}_{++}^{K-1}$, the corresponding composition is obtained by $\varphi_{\text{comp}}^{-1}(o) = (o_1^2, \ldots, o_K^2)^\top := o^{\circ 2}$. The estimation methods are obtained by applying the general constructions above to the embedded outcome $O_{i,t} = \varphi_{\text{comp}}(X_{i,t})$.

An alternative approach in this setting is compositional DID, denoted by CoDiD Boussim (2026). CoDiD imposes a componentwise multiplicative untreated trend. This operation has no direct analogue for a general manifold-valued outcome. Let $M_{a,t,k} > 0$ denote the aggregate amount in category k for group $a \in \{T, C\}$ and period $t \in \{0,1\}$, and define the aggregate composition

$$\bar{x}_{a,t,k} := \frac{M_{a,t,k}}{\sum_{\ell=1}^K M_{a,t,\ell}}, \qquad k = 1, \ldots, K.$$

The CoDiD counterfactual composition for the treated group is constructed componentwise as

$$\widetilde{x}_{T,1,k}^{\text{CoDiD}}(0) = \bar{x}_{T,0,k} \frac{\bar{x}_{C,1,k}}{\bar{x}_{C,0,k}}, \qquad k = 1, \ldots, K,$$

and then closed back to the simplex by normalization: $\widehat{x}_{T,1,k}^{\text{CoDiD}}(0) = \frac{\widetilde{x}_{T,1,k}^{\text{CoDiD}}(0)}{\sum_{\ell=1}^K \widetilde{x}_{T,1,\ell}^{\text{CoDiD}}(0)}, k = 1, \ldots, K.$

6.2. **Data-generating process.** This subsection specifies the baseline compositions, the untreated transition, and the post-period shocks of the simulation. The simulation imposes no treatment effect. It evaluates how well the estimators recover the untreated post-period outcomes of the treated group, which are generated by the common untreated transition map and the post-period shocks defined below. The three categories are denoted by A, B, and U, where A and B are the two major parties and U is the residual category of other candidates. We write a simulated vote-share vector as $x = (x_A, x_B, x_U)^\top \in \Delta_{\mathrm{v}}^\circ$, where $\Delta_{\mathrm{v}}^\circ := \{x \in (0,1)^3 : x_A + x_B + x_U = 1\}$. All simulated vote-share vectors are embedded into $\mathbb{S}^2$ using the square-root map φ_{comp} from Section 6.1 with $K = 3$. Thus the simulated manifold-valued outcome is $O_{i,a,t}^{\mathrm{sim}}(z) = \varphi_{\mathrm{comp}}\{x_{i,a,t}^{\mathrm{sim}}(z)\} \in \mathbb{S}_{++}^2$, for unit $i = 1, \ldots, n_a$, group $a \in \{C, T\}$, period $t \in \{0, 1\}$, and treatment state $z \in \{0, 1\}$. Reported vote-share summaries are obtained by applying $\varphi_{\mathrm{comp}}^{-1}$ to the corresponding sphere-valued summaries.

The baseline composition follows a finite mixture of logistic-normal components. Let $L_{i,a} \in \{1, 2, 3\}$ be a latent electoral class with group-specific probabilities $\mathbb{P}(L_{i,a} = k) = \omega_{a,k}$. Conditional on $L_{i,a} = k$, draw a two-dimensional latent logit vector $\zeta_{i,a} \mid L_{i,a} = k \sim N_2(b_k, V_{\mathrm{logit}})$ with $V_{\mathrm{logit}} = 0.20 I_2$. For $u = (u_1, u_2)^\top \in \mathbb{R}^2$, define the multinomial-logit map $\mathrm{soft}_{\mathrm{v}}(u) := \{1 + \exp(u_1) + \exp(u_2)\}^{-1}(\exp(u_1), \exp(u_2), 1)^\top$. The pre-treatment composition is $x_{i,a,0}^{\mathrm{sim}} = \mathrm{soft}_{\mathrm{v}}(\zeta_{i,a})$, so the latent logit vector $\zeta_{i,a}$ determines the choice probabilities for Party A, Party B, and an outside option whose utility is normalized to zero. The corresponding manifold-valued pre-treatment outcome is $O_{i,a,0}^{\mathrm{sim}}(0) = \varphi_{\mathrm{comp}}(x_{i,a,0}^{\mathrm{sim}})$. Since treatment has not yet occurred at $t = 0$, the observed pre-treatment outcome is $O_{i,a,0}^{\mathrm{sim}} = O_{i,a,0}^{\mathrm{sim}}(0)$ for $a \in \{C, T\}$. Table 1 lists the class means b_k, the class probabilities, and the expected vote shares of each class.

The different class probabilities create baseline compositional imbalance between the treated and control groups. Class 1 represents a balanced population between the two major parties, whereas Class 2 and Class 3 are tilted toward Party A and Party B, respectively, with the dominant party leading by approximately 21.7 percentage points in expected vote share.

The untreated temporal evolution combines the common untreated transition map, a nonlinear map on the raw composition scale, with independent post-period shocks introduced below. We denote this map by $\mathcal{T}_\kappa^{(0)}$, where the transition intensity κ is the fraction of the distance to each of two target shares that the map closes between the two periods. Applied to a vote-share vector, $\mathcal{T}_\kappa^{(0)}$ first moves the share of the other candidates a fraction κ of the way toward 0.10. It then divides the remaining share between the two major parties so that the share of Party A in their combined

Class k	b_k	$\omega_{C,k}$	$\omega_{T,k}$	Party A	Party B	Other
1	$(2.50, 2.50)^\top$	0.45	0.35	48.03	48.03	3.94
2	$(2.70, 2.20)^\top$	0.30	0.55	58.83	37.14	4.03
3	$(2.20, 2.70)^\top$	0.25	0.10	37.14	58.83	4.03

TABLE 1. Latent electoral classes in the simulation design. For each class k, the table reports the mean b_k of the latent logit vector, the class probabilities $\omega_{C,k}$ and $\omega_{T,k}$ in the control and treated groups, and the expected vote shares of Party A, Party B, and the other candidates, in percent and rounded to two decimal places, under $V_{\text{logit}} = 0.20 I_2$.

vote moves a fraction κ of the way toward 0.52. That is, for $x \in \Delta_{\text{v}}^{\circ}$,

$$\begin{aligned}
\mathcal{T}_\kappa^{(0)}(x)_U &= (1-\kappa)x_U + 0.10\kappa, \\
\mathcal{T}_\kappa^{(0)}(x)_A &= \{1 - \mathcal{T}_\kappa^{(0)}(x)_U\}\Big\{(1-\kappa)\frac{x_A}{x_A + x_B} + 0.52\kappa\Big\}, \\
\mathcal{T}_\kappa^{(0)}(x)_B &= \{1 - \mathcal{T}_\kappa^{(0)}(x)_U\}\Big\{(1-\kappa)\frac{x_B}{x_A + x_B} + 0.48\kappa\Big\}.
\end{aligned}$$

The first target represents a moderate pull of the residual category toward a common level. The second target breaks exact two-party symmetry and induces a mild regression-to-the-mean force within the two-party share. We use $\kappa \in \{0, 0.05, 0.10, 0.15, 0.20\}$. The value $\kappa = 0$ gives the identity map, whereas larger values of κ generate stronger common regression toward the two targets. The map is common across groups but nonlinear in the original simplex coordinates.

Idiosyncratic post-period variation is added by a multiplicative logistic-normal perturbation. For any $x \in \Delta_{\text{v}}^{\circ}$ and $\varepsilon = (\varepsilon_A, \varepsilon_B, \varepsilon_U)^\top \in \mathbb{R}^3$, define $\mathcal{E}_{\sigma_{\text{sim}}}(x;\varepsilon)_j = x_j \exp(\varepsilon_j)/\{\sum_{\ell \in \{A,B,U\}} x_\ell \exp(\varepsilon_\ell)\}$ for $j \in \{A, B, U\}$. The shocks are mutually independent, satisfy $\varepsilon_{i,a,1,j}^{(0)} \overset{\text{iid}}{\sim} N(0, \sigma_{\text{sim}}^2)$ with $\sigma_{\text{sim}} = 0.1$, and are also independent of latent electoral classes and pre-treatment compositions. For both groups, the untreated post-period composition is

$$x_{i,a,1}^{\text{sim}}(0) = \mathcal{E}_{\sigma_{\text{sim}}}\left[\mathcal{T}_\kappa^{(0)}\{x_{i,a,0}^{\text{sim}}\}; \varepsilon_{i,a,1}^{(0)}\right],$$

and the corresponding outcome is $O_{i,a,1}^{\text{sim}}(0) = \varphi_{\text{comp}}\{x_{i,a,1}^{\text{sim}}(0)\}$. For the control group, the untreated post-period outcome is observed, $O_{i,C,1}^{\text{sim}} = O_{i,C,1}^{\text{sim}}(0)$. For the treated group, $O_{i,T,1}^{\text{sim}}(0)$ is generated only to construct the oracle benchmark and is not used by the estimators. No treatment effect is imposed, because the target is the untreated counterfactual of the treated group.

6.3. **Results.** This subsection compares the estimators by their Fréchet MSE and reports the rejection rates of the pretrend tests. The estimators are compared using the finite-sample oracle Fréchet mean $\widehat{\mu}^{\mathrm{or}}_{T,1}(0) \in \arg\min_{q\in\mathcal{M}_{\mathrm{comp}}} n_T^{-1}\sum_{i=1}^{n_T} d_{\mathcal{M}_{\mathrm{comp}}}\{O^{\mathrm{sim}}_{i,T,1}(0), q\}^2$. If an estimator returns a composition-valued summary $\widehat{x}_{T,1}(0)$, it is evaluated on the manifold as $\widehat{\mu}_{T,1}(0) = \varphi_{\mathrm{comp}}\{\widehat{x}_{T,1}(0)\}$. The Monte Carlo loss is $d^2_{\mathcal{M}_{\mathrm{comp}}}\{\widehat{\mu}_{T,1}(0), \widehat{\mu}^{\mathrm{or}}_{T,1}(0)\}$. For distribution-valued estimators, including Gaussian OT and 1NN OT, the quantity $\widehat{\mu}_{T,1}(0)$ is obtained by averaging the predicted counterfactual points in the common chart and mapping this chart mean back to $\mathcal{M}_{\mathrm{comp}}$ by Ψ^{-1}_{ref}. For these estimators, the loss compares a chart-mean summary with the oracle sample Fréchet mean. Thus the reported Fréchet MSE is computed in three steps in each Monte Carlo replication: first construct the oracle untreated treated-post Fréchet mean from the generated but unobserved $O^{\mathrm{sim}}_{i,T,1}(0)$; second, convert each estimator's counterfactual output to its manifold-valued summary as described above; third, record the squared spherical geodesic distance between the estimator's summary and the oracle.

We compare seven estimators. CoDiD is the composition-specific benchmark described in Section 6.1. GDiD and TDiD impose the geodesic and tangent-space mean parallel-trend restrictions, respectively. Gaussian OT (Time) and 1NN OT (Time) estimate the common untreated transition map from the control group's transition from period 0 to period 1 and apply it to the treated baseline distribution. Gaussian OT (State) and 1NN OT (State) instead learn a cross-group baseline map from control to treated units and then apply that map to the control post-period distribution.

Gaussian OT (Time) and Gaussian OT (State) approximate the source and target samples in the common chart by Gaussian laws with the corresponding sample means and with the sample covariance matrices plus $10^{-6}I_d$. The 1NN OT rows use the 1NN barycentric extension of Section 5.2. The two samples in each empirical transport problem have the same size and uniform weights, so an optimal plan can be taken to be an assignment that minimizes the sum of squared distances, and we compute it exactly by the Hungarian algorithm. The simulation has $B = 1000$ replications for each design. Other extension choices remain sensitivity specifications.

Table 2 reports the Fréchet MSE, the average over the replications of the Monte Carlo loss, multiplied by 10^5, for each estimator and transition intensity, separately for $n = 150$ and $n = 300$ observations per group. It shows that the estimators based on the common untreated transition map are stable across the values of κ. This is consistent with the construction of the DGP, which applies the common untreated transition map to every unit. In contrast, GDiD, TDiD, and the state-map estimators rely on structures that are not built into the DGP. Their MSE increases as

$n = 150$					
Estimator	$\kappa = 0.00$	$\kappa = 0.05$	$\kappa = 0.10$	$\kappa = 0.15$	$\kappa = 0.20$
CoDiD	2.126	1.705	2.879	5.115	8.965
GDiD	1.636	2.449	5.224	8.769	14.147
TDiD	1.633	2.441	5.212	8.760	14.146
Gaussian OT (Time)	1.795	1.786	1.918	1.936	1.987
Gaussian OT (State)	1.621	2.417	5.298	8.820	14.248
1NN OT (Time)	2.244	2.182	2.240	2.187	2.282
1NN OT (State)	2.087	3.237	7.301	12.621	20.799
$n = 300$					
Estimator	$\kappa = 0.00$	$\kappa = 0.05$	$\kappa = 0.10$	$\kappa = 0.15$	$\kappa = 0.20$
CoDiD	1.353	0.835	1.788	3.922	7.628
GDiD	0.826	1.609	3.898	7.419	12.725
TDiD	0.824	1.603	3.889	7.412	12.726
Gaussian OT (Time)	0.940	0.898	0.989	1.014	1.056
Gaussian OT (State)	0.808	1.576	3.846	7.332	12.607
1NN OT (Time)	1.041	1.017	1.084	1.076	1.103
1NN OT (State)	0.925	2.054	5.208	10.055	16.956

TABLE 2. Fréchet MSE, multiplied by 10^5, by estimator, transition intensity, and per-group sample size.

κ grows, because stronger nonlinear temporal movement makes the induced mean-level geodesic or tangent-space parallel trends increasingly inaccurate. The state-map estimators are especially sensitive, since the DGP does not contain a common map from control units to treated units. For 1NN OT (Time) at $n = 300$, the scaled MSE ranges from 1.017 to 1.103 over the five transition intensities; the corresponding MSE of 1NN OT (State) rises from 0.925 to 16.956.

We further examine the extent to which the mean parallel-trend restrictions imposed by GDiD and TDiD are compatible with the simulated data-generating process. For each value of κ and each per-group sample size $n \in \{150, 300\}$, we repeat the simulation $B = 1000$ times and report the empirical rejection probability of the corresponding pretrend test at nominal level $\alpha = 0.05$. The diagnostic is implemented using the same periods, $t = 0$ and $t = 1$, that are used by the estimators.

n	Test	$\kappa = 0.00$	$\kappa = 0.05$	$\kappa = 0.10$	$\kappa = 0.15$	$\kappa = 0.20$
150	GDiD	0.050	0.094	0.181	0.262	0.325
	TDiD	0.050	0.093	0.181	0.261	0.326
300	GDiD	0.049	0.128	0.311	0.478	0.600
	TDiD	0.049	0.127	0.311	0.477	0.603

TABLE 3. Monte Carlo empirical rejection rates at nominal level $\alpha = 0.05$ for the pretrend tests, by transition intensity κ and per-group sample size. The number of Monte Carlo replications is $B = 1000$.

Table 3 reports the resulting rejection rates. When $\kappa = 0$, the rejection probabilities are close to the nominal level for both tests and both sample sizes in this design. As κ increases, the rejection rates rise monotonically. This pattern shows that stronger nonlinear temporal movement produces larger departures from the mean parallel-trend restrictions underlying GDiD and TDiD in this design.

7. Empirical analysis: early voting and county-level vote shares

This section illustrates the estimators with county-level presidential election returns. Section 7.1 describes the data and comparison groups, Section 7.2 examines observed movements and pretreatment diagnostics, Section 7.3 describes the counterfactual estimation and its uncertainty, and Section 7.4 reports the estimated effects.

7.1. **Data and comparison groups.**

We use county-level presidential election returns from the MIT Election Data and Science Lab (MEDSL), distributed through Harvard Dataverse as MIT Election Data and Science Lab (2018). We retain the 2000, 2004, and 2008 elections and records for the office of U.S. President.

We use the four-state comparison in Boussim (2026), assigning Maryland and New Jersey to the treated group and New York and Pennsylvania to the control group. The main comparison uses 2004 as the baseline and 2008 as the post-treatment election. The 2000 election provides an additional pretreatment observation. Early voting is understood broadly to include voting before Election Day in person or by mail. The outcomes measure all presidential votes recorded in a county, so the application concerns the composition of total election returns under this policy comparison.

Year	Group	Counties	Democrat	Republican	Other
2000	Control	129	44.741	51.168	4.091
2000	Treated	45	49.79	46.83	3.38
2004	Control	129	43.425	53.349	3.226
2004	Treated	45	47.260	51.726	1.014
2008	Control	129	48.424	50.184	1.392
2008	Treated	45	52.25	46.73	1.02

TABLE 4. Arithmetic county mean vote shares, in percent, for the MEDSL-only sample. Counties receive equal weight within each group. Percentages are rounded to two or three decimal places so that each displayed row sums to 100.

We collapse the reported party labels into Democrat, Republican, and Other. The analysis uses only the MEDSL file and keeps its reported party labels throughout. Each county-year composition is obtained by dividing the three category counts by their sum. We retain counties observed in all three elections with positive total valid votes, yielding 174 counties and 522 county-year observations. The treated group contains 45 counties, comprising 24 in Maryland and 21 in New Jersey. The control group contains 129 counties, comprising 62 in New York and 67 in Pennsylvania. Every retained composition has three positive components. Table 4 summarizes the county compositions.

7.2. **Observed movements and pretreatment diagnostics.** We examine group means and county trajectories to describe the observed movements before estimating counterfactual outcomes. Figure 11 plots the arithmetic county means of the three recorded shares. Between 2004 and 2008, the control group's mean Other share falls from 3.226% to 1.392%, while the treated group's mean remains close to 1%. Thus, the recorded residual category has a substantial group-specific movement over the main comparison period.

Figure 12 shows the movement of each county in the Democrat-Republican share plane; the Other share is the remaining percentage. Figure 13 shows the average movements within ten fixed bins of counties per group. The binned trajectories reveal heterogeneity within each group and different movements across the two periods. In particular, many control counties have high recorded Other shares in 2004 and substantially lower shares in 2008, which appears in panel (b) as movement toward the dashed line. These patterns motivate examining how the estimated counterfactual changes when the model uses the distribution of county compositions.

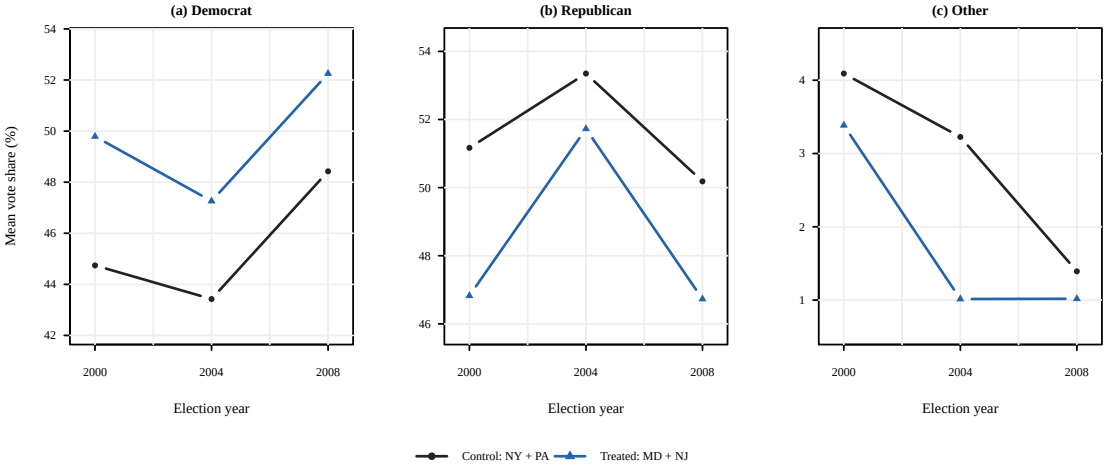


FIGURE 11. Arithmetic county mean shares in the three presidential elections. Black circles show the 129 control counties in New York and Pennsylvania, and blue triangles show the 45 treated counties in Maryland and New Jersey. The panels display Democrat, Republican, and Other shares on separate vertical scales, all in percent. All values use the MEDSL classifications.

Test	Wald statistic	p-value	Decision
GDiD	72.795	1.56×10^{-16}	Reject
TDiD	72.059	2.25×10^{-16}	Reject

TABLE 5. Pretreatment diagnostics for the 2000-to-2004 transition. Decisions use a nominal significance level of 5%. Both tests are computed from the paired county observations, with the reference point at the sample Fréchet mean of the pooled 2000 observations.

We apply the GDiD and TDiD Wald tests of Section 4 to the 2000-to-2004 pretreatment transition. Both tests reject their respective mean parallel-trend restrictions at the 5% level, as reported in Table 5. We therefore retain GDiD and TDiD as benchmark specifications whose corresponding pretreatment restrictions are rejected in these data.

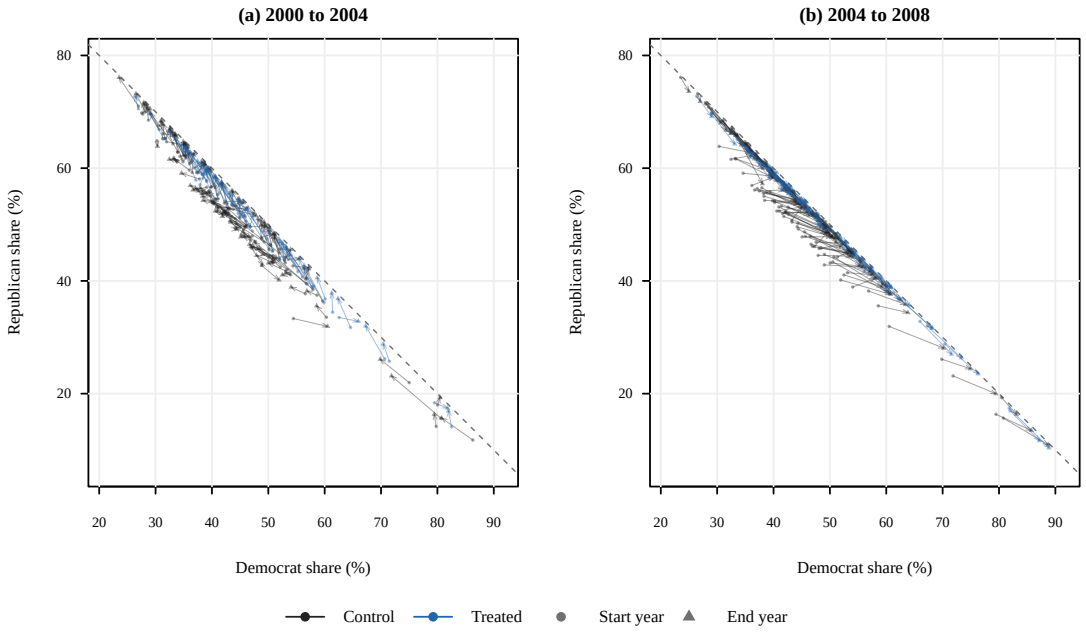


FIGURE 12. Observed county trajectories in the Democrat-Republican share plane from 2000 to 2004 in panel (a) and from 2004 to 2008 in panel (b). Each arrow connects the two observed compositions of the same county. Black denotes the control group and blue the treated group; circles mark the start year and triangles the end year. Both axes are percentages, and the dashed line marks the points at which the Democrat and Republican shares sum to 100%, so the vertical distance from a point to this line is its Other share.

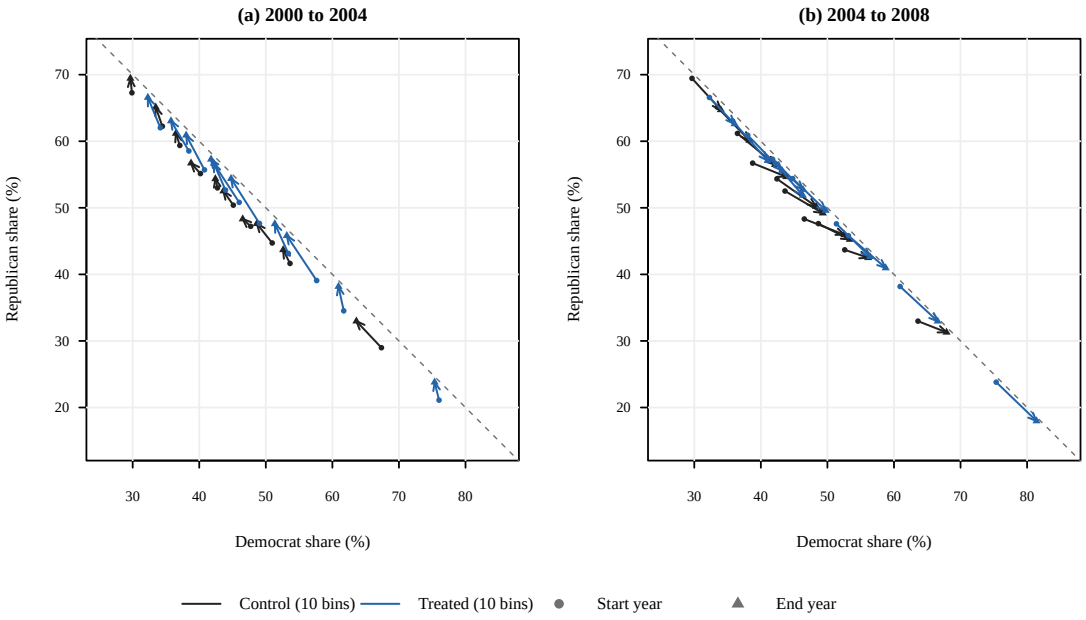


FIGURE 13. Mean county trajectories in the Democrat-Republican share plane. Within each group, all counties are sorted by their 2000 Democrat share and divided into ten bins of nearly equal size, held fixed across elections. Each arrow connects the arithmetic county means for one bin from 2000 to 2004 in panel (a) and from 2004 to 2008 in panel (b). Black denotes the control group and blue the treated group; circles mark the start year and triangles the end year. Both axes are percentages, and the dashed line marks the points at which the Democrat and Republican shares sum to 100%, so the vertical distance from a point to this line is its mean Other share.

7.3. **Counterfactual estimation and uncertainty.** We estimate the treated group's composition in 2008 under the untreated counterfactual with the seven estimators of Section 6.3 and compare it with the observed 2008 composition. The reported effect for each category is the observed composition summary minus the corresponding estimated counterfactual summary, expressed in percentage points. The three effects sum to zero before rounding.

We form 95% percentile intervals using 500 bootstrap replications. Counties are sampled with replacement within the treated and control groups, retaining each sampled county's three-election trajectory. The chart reference is recomputed in each replication.

7.4. **Estimated effects.** Table 6 reports the estimates. CoDiD, GDiD, and TDiD give positive effects on Other of approximately 0.51 to 0.54 percentage points, with intervals above zero. By contrast, 1NN OT using the time map gives an Other effect of -0.293 percentage points, with a 95% interval of $[-0.483, -0.046]$. Gaussian OT using the time map also gives a negative point estimate, -0.151, but its interval includes zero. Both state-map OT intervals for Other include zero.

All Democratic-share intervals include zero. The Republican-share intervals of CoDiD, GDiD, and TDiD lie below zero, whereas the Republican-share intervals of the OT specifications include zero. Thus, the clearest disagreement across the specifications concerns the direction of the effect on the recorded Other category.

Method	Democrat	Republican	Other
CoDiD	1.034 [-0.037, 2.122]	-1.543 [-2.624, -0.447]	0.509 [0.288, 0.765]
GDiD	0.711 [-0.018, 1.474]	-1.254 [-1.977, -0.509]	0.543 [0.297, 0.796]
TDiD	0.715 [-0.010, 1.475]	-1.259 [-1.983, -0.514]	0.544 [0.297, 0.799]
Gaussian OT (Time)	0.433 [-0.353, 1.222]	-0.282 [-1.059, 0.489]	-0.151 [-0.345, 0.078]
Gaussian OT (State)	-0.150 [-1.990, 1.450]	0.124 [-1.511, 1.933]	0.025 [-0.150, 0.220]
1NN OT (Time)	0.969 [-0.327, 2.719]	-0.676 [-2.291, 0.628]	-0.293 [-0.483, -0.046]
1NN OT (State)	-0.188 [-3.236, 2.240]	0.177 [-2.252, 3.261]	0.011 [-0.166, 0.228]

TABLE 6. Effects on the three recorded vote shares, in percentage points. Brackets contain 95% percentile intervals from the 500 paired-county bootstrap replications. CoDiD uses aggregate vote totals, GDiD uses a Fréchet mean, and TDiD and OT use the common-chart mean for both observed and counterfactual summaries.

8. Conclusion

This paper has studied difference-in-differences for outcomes that take values in a Riemannian manifold, such as vote-share compositions. GDiD, TDiD, and CoDiD restrict an average movement of the outcome, so the average movements of the treated and control groups must agree in both size and direction. When the untreated movement varies with the baseline position and the two groups start from different positions, the counterfactual implied by such a restriction can point in a counterintuitive direction.

MCiC instead targets the untreated counterfactual distribution of the treated group. It assumes a common untreated transition map, which moves each baseline position on a common tangent-space chart to its untreated post-treatment position in the same way for the treated and control groups. Under the conditions of Section 5.1, the optimal transport between the pre-treatment and post-treatment distributions of the control group identifies this map, and applying the map to the baseline distribution of the treated group recovers the counterfactual distribution. For estimation, we proposed 1NN OT and derived a general-dimensional upper bound on its convergence rate. In the simulations, the performance of 1NN OT (Time) is stable over the range of transition intensities that we examine, whereas the errors of GDiD and TDiD grow as the untreated movement toward a central composition becomes stronger. In the early-voting application, 1NN OT (Time) gives a negative estimated effect on the share of the residual Other category, whereas CoDiD, GDiD, and TDiD give positive estimated effects.

Appendix A. Proofs for Section 3

A.1. **Proof of Theorem 3.1.**

Proof. The proof first identifies the three observed Fréchet means, then constructs the GDiD point from them and the known Riemannian structure, and finally uses the geodesic parallel trends condition in Assumption 3.1 (3) to show that this point equals $\mu_{T,1}(0)$.

By the observation rule, $O_{C,0} = O_{C,0}(0)$, $O_{C,1} = O_{C,1}(0)$, and $O_{T,0} = O_{T,0}(0)$. Hence, the laws of $O_{C,0}$, $O_{C,1}$, and $O_{T,0}$ are identified from the observed distribution. Since $d_{\mathcal{M}}$ is known, the corresponding Fréchet functionals $F_{C,0}$, $F_{C,1}$, and $F_{T,0}$ are also identified. By Assumption 3.1 (1), these functionals have unique minimizers. Therefore, $\mu_{C,0}$, $\mu_{C,1}$, and $\mu_{T,0}$ are identified as functions of the observed distribution.

By Assumption 3.1 (2), the points $\mu_{C,0}$, $\mu_{C,1}$, and $\mu_{T,0}$ lie in the same open strongly convex neighborhood. Hence, the logarithmic vector $\mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})$ and the unique minimizing geodesic $\eta_{C\to T}$ from $\mu_{C,0}$ to $\mu_{T,0}$ are well-defined. The Levi–Civita connection then defines a unique parallel transport map $\Pi_{\eta_{C\to T}} : T_{\mu_{C,0}}\mathcal{M} \to T_{\mu_{T,0}}\mathcal{M}$ along this geodesic. Therefore, $v_{\mathrm{GDiD}} := \Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1}) \in T_{\mu_{T,0}}\mathcal{M}$ is uniquely determined by the identified Fréchet means and the known Riemannian structure.

By Assumption 2.1 and the Hopf–Rinow theorem, $\mathrm{Exp}_{\mu_{T,0}}$ is defined on the entire tangent space $T_{\mu_{T,0}}\mathcal{M}$. Therefore, the point

$$\mu_{T,1}^{\mathrm{GDiD}}(0) := \mathrm{Exp}_{\mu_{T,0}}(v_{\mathrm{GDiD}}) = \mathrm{Exp}_{\mu_{T,0}}\{\Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})\}$$

is well-defined and is constructed only from identified quantities and the known Riemannian structure. Thus, $\mu_{T,1}^{\mathrm{GDiD}}(0)$ is identified.

Moreover, by the definition of the geodesic transport map in the setup, $\Gamma_{\mu_{C,0},\mu_{C,1}}(\mu_{T,0}) = \mathrm{Exp}_{\mu_{T,0}}\{\Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1})\}$. Hence, $\mu_{T,1}^{\mathrm{GDiD}}(0) = \Gamma_{\mu_{C,0},\mu_{C,1}}(\mu_{T,0})$.

Finally, Assumption 3.1 (3) states that $\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\} = \Pi_{\eta_{C\to T}} \mathrm{Log}_{\mu_{C,0}}(\mu_{C,1}) = v_{\mathrm{GDiD}}$. Therefore, $\mu_{T,1}^{\mathrm{GDiD}}(0) = \mathrm{Exp}_{\mu_{T,0}}\{\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\}\}$. Since $\mu_{T,0}$ and $\mu_{T,1}(0)$ lie in the same strongly convex neighborhood by Assumption 3.1 (2), the logarithmic map $\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\}$ is uniquely defined, and the local inverse relation between Exp and Log gives $\mathrm{Exp}_{\mu_{T,0}}\{\mathrm{Log}_{\mu_{T,0}}\{\mu_{T,1}(0)\}\} = \mu_{T,1}(0)$. Hence, $\mu_{T,1}^{\mathrm{GDiD}}(0) = \mu_{T,1}(0)$. □

A.2. **Proof of Theorem 3.2.**

Proof. The proof identifies the three observed chart means, uses the tangent-space parallel trends condition in Assumption 3.2 (1) to express the counterfactual chart mean through them, and then maps this chart mean back to $\mathcal{M}$.

By the observation rule, $O_{C,0} = O_{C,0}(0)$, $O_{C,1} = O_{C,1}(0)$, and $O_{T,0} = O_{T,0}(0)$. Therefore, the laws $P_{C,0}$, $P_{C,1}$, and $P_{T,0}$ are identified from the observed distribution. Since Ψ_{ref} is known, the laws of $Y_{C,0}$, $Y_{C,1}$, and $Y_{T,0}$ are identified as the pushforwards of these observed laws under Ψ_{ref}. Hence the chart means $m_{C,0}^{Y}$, $m_{C,1}^{Y}$, and $m_{T,0}^{Y}$ are identified.

The counterfactual chart mean $m_{T,1}^{Y,(0)} = \mathbb{E}[Y_{T,1}(0)]$ is not directly observed. However, Assumption 3.2 (1) implies $m_{T,1}^{Y,(0)} - m_{T,0}^{Y} = m_{C,1}^{Y} - m_{C,0}^{Y}$. Therefore, $m_{T,1}^{Y,(0)} = m_{T,0}^{Y} + \left(m_{C,1}^{Y} - m_{C,0}^{Y}\right)$. The right-hand side consists only of identified chart means, so $m_{T,1}^{Y,(0)}$ is identified.

Finally, Assumption 3.2 (2) ensures that this identified chart mean lies in $\mathcal{Y}_{\mathrm{TDiD}}$. Since $\mathcal{U}_{\mathrm{TDiD}} \subset \mathcal{U}_{\mathrm{ref}}$ and Ψ_{ref} is a chart on $\mathcal{U}_{\mathrm{ref}}$, the inverse map Ψ_{ref}^{-1} is well-defined at this point. Thus $\theta_{T,1}^{\mathrm{TDiD}}(0) = \Psi_{\mathrm{ref}}^{-1}\left\{m_{T,0}^{Y} + \left(m_{C,1}^{Y} - m_{C,0}^{Y}\right)\right\}$ is identified. The equivalent expression follows from the setup identity $\Psi_{\mathrm{ref}}^{-1}(y) = \mathrm{Exp}_{\mu_{\mathrm{ref}}}(B_{\mathrm{ref}} y)$. □

Appendix B. Proofs for Section 4

B.1. **Proof of Lemma 4.1.**

Proof. The goal is the joint chart limit

$$\sqrt{n}\{\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) - \Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\} \Rightarrow \mathcal{N}\Big(0, \mathrm{diag}\{\rho_T^{-1}\Omega_T, \rho_C^{-1}\Omega_C\}\Big), \qquad \Omega_a = A_a C_a A_a^{\top},$$

with the score covariance C_a and block inverse-Hessian matrix A_a defined below.

The proof has three steps. It first shows that the sample Fréchet means are consistent, so that they lie in $\mathcal{U}_{\mathrm{PG}}$ with probability tending to one. It then derives an asymptotic linear representation in the chart Ψ_{PG} for each group and period. Finally, it stacks the two periods within each group and combines the two independent groups.

The four population and sample means are ordered as $\mathbf{m}_{\mathrm{G}} = (\mu_{T,s_0}, \mu_{T,s_1}, \mu_{C,s_0}, \mu_{C,s_1})$ and $\widehat{\mathbf{m}}_{\mathrm{G}} = (\widehat{\mu}_{T,s_0}, \widehat{\mu}_{T,s_1}, \widehat{\mu}_{C,s_0}, \widehat{\mu}_{C,s_1})$. On $\mathcal{U}_{\mathrm{PG}}^4$, the product chart used in the conclusion is

$$\Phi_{\mathrm{G}}(m_{T,s_0}, m_{T,s_1}, m_{C,s_0}, m_{C,s_1}) = (\Psi_{\mathrm{PG}}(m_{T,s_0}), \Psi_{\mathrm{PG}}(m_{T,s_1}), \Psi_{\mathrm{PG}}(m_{C,s_0}), \Psi_{\mathrm{PG}}(m_{C,s_1})).$$

As stipulated in Lemma 4.1, $\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})$ equals a fixed deterministic vector in $\mathbb{R}^{4d}$ if any sample mean lies outside this chart. Write $n := n_T + n_C$; Assumption 4.1 gives $n_a/n \to \rho_a \in (0,1)$ for $a \in \{T, C\}$.

Fix a group $a \in \{T, C\}$. For $j \in \{0, 1\}$, let $P_{a,j}$ be the law of O_{i,a,s_j} and recall that $F_{a,j}(q) := \mathbb{E}\left[d_{\mathcal{M}}(O_{a,s_j}, q)^2\right]$ for $q \in \mathcal{M}$ has a unique minimizer μ_{a,s_j} by Assumption 4.2(1). Let $u_{a,j} := \Psi_{\mathrm{PG}}(\mu_{a,s_j}) \in \mathbb{R}^d$.

For each $j \in \{0, 1\}$, consider the $\mathcal{M}$-valued sample $O_{1,a,s_j}, \ldots, O_{n_a,a,s_j}$. By Assumption 4.1, this sample is i.i.d. within group a, and $n_a \to \infty$. We first establish consistency without using any CLT. By Assumption 2.1 and the Hopf–Rinow theorem, every closed and bounded subset of $(\mathcal{M}, d_{\mathcal{M}})$ is compact. The law $P_{a,j}$ has compact support by Assumption 4.2(3), so $F_{a,j}$ is finite on $\mathcal{M}$, and $F_{a,j}$ has the unique minimizer μ_{a,s_j} by Assumption 4.2(1). Theorem 2.3 of Bhattacharya and Patrangenaru (2003) therefore shows that every measurable selection $\widehat{\mu}_{a,s_j}$ from the sample Fréchet mean set converges to μ_{a,s_j} almost surely. This holds for both groups and both periods.

Since $\mathcal{U}_{\mathrm{PG}}$ is open and contains the four population means by Assumption 4.2(2), the event $\mathcal{E}_n^{\mathrm{G}}$ of Section 4.1, on which all four sample Fréchet means lie in $\mathcal{U}_{\mathrm{PG}}$, satisfies $\mathbb{P}(\mathcal{E}_n^{\mathrm{G}}) \to 1$.

On $\mathcal{E}_n^{\mathrm{G}}$, the vector $\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})$ is given by the product chart, and on $(\mathcal{E}_n^{\mathrm{G}})^c$ it equals the fixed vector in Lemma 4.1. Changing random vectors on events whose probabilities tend to zero does not affect $o_p(1)$ remainders or limits in distribution. It therefore suffices to derive the asymptotic linear representation below on $\mathcal{E}_n^{\mathrm{G}}$, where $\Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_j})$ is defined. We work in the chart Ψ_{PG} and write $F_{a,j}^{\Psi_{\mathrm{PG}}}(u) := F_{a,j}\{\Psi_{\mathrm{PG}}^{-1}(u)\}$. By Assumption 4.2(5), there is a neighborhood of $u_{a,j}$ on which $u \mapsto d_{\mathcal{M}}\{o, \Psi_{\mathrm{PG}}^{-1}(u)\}^2$ is twice continuously differentiable for every $o \in \mathrm{supp}(P_{a,j})$, with first and second derivatives that are jointly continuous and bounded, and the Hessian $H_{a,j} := D^2 F_{a,j}^{\Psi_{\mathrm{PG}}}(u_{a,j})$ is positive definite.

To obtain the joint limit for the two pre-treatment periods, we use the asymptotic linear representation behind Theorem 2.1 of Bhattacharya and Patrangenaru (2005). Write $h_{a,j}(o,u) := d_{\mathcal{M}}\{o, \Psi_{\mathrm{PG}}^{-1}(u)\}^2$ and $\psi_{a,j}(o;u) := -D_u h_{a,j}(o,u) \in \mathbb{R}^d$. On the neighborhood above, the bounds of Assumption 4.2(5) give their integrability conditions (2.9) and, by dominated convergence, their continuity condition (2.10). The same bounds allow differentiation under the expectation, so $\mathbb{E}\{\psi_{a,j}(O_{a,s_j}; u_{a,j})\}$ is the negative gradient of $F_{a,j}^{\Psi_{\mathrm{PG}}}$ at its minimizer $u_{a,j}$, which is zero, and their matrix (2.12) equals $H_{a,j}$. Since the scores are also bounded, $n_a^{-1/2}\sum_{i=1}^{n_a} \psi_{a,j}(O_{i,a,s_j}; u_{a,j}) = O_p(1)$. By consistency, $\Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_j})$ lies in the neighborhood with probability tending to one, so their expansion (2.7)–(2.12) applies, and solving their equation (2.11), whose score (2.5) is $-\psi_{a,j}$, gives, for each $j \in \{0,1\}$,

$$\sqrt{n_a}\left\{\Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_j}) - u_{a,j}\right\} = H_{a,j}^{-1}\frac{1}{\sqrt{n_a}}\sum_{i=1}^{n_a}\psi_{a,j}(O_{i,a,s_j}; u_{a,j}) + o_p(1).$$

Define the stacked score

$$\xi_{i,a} := \begin{pmatrix} \psi_{a,0}(O_{i,a,s_0}; u_{a,0}) \\ \psi_{a,1}(O_{i,a,s_1}; u_{a,1}) \end{pmatrix} \in \mathbb{R}^{2d}.$$

Since the paired pre-period trajectories $\mathbf{O}_{i,a} = (O_{i,a,s_0}, O_{i,a,s_1})$ are i.i.d. by Assumption 4.1, the vectors $\xi_{i,a}$ are i.i.d. The dependence between O_{i,a,s_0} and O_{i,a,s_1} for the same individual is fully captured by the off-diagonal covariance blocks of $\xi_{i,a}$ and does not affect the joint CLT.

Each stacked score is bounded, because Assumption 4.2(5) bounds the first derivatives of the squared distance uniformly over the support, so $\xi_{i,a}$ has a finite covariance matrix. It also has mean zero, because each of its components has mean zero, as shown above.

Let $C_a := \mathrm{Var}(\xi_{i,a}) \in \mathbb{R}^{2d \times 2d}$ and $A_a := \mathrm{diag}\left(H_{a,0}^{-1}, H_{a,1}^{-1}\right)$. Stacking the two asymptotic linear representations yields

$$\begin{aligned}
&\sqrt{n_a}\left[\begin{pmatrix}\Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_0})\\ \Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_1})\end{pmatrix} - \begin{pmatrix}u_{a,0}\\ u_{a,1}\end{pmatrix}\right] \\
&\quad = A_a \frac{1}{\sqrt{n_a}} \sum_{i=1}^{n_a} \xi_{i,a} + o_p(1).
\end{aligned}$$

By the multivariate central limit theorem, $\frac{1}{\sqrt{n_a}} \sum_{i=1}^{n_a} \xi_{i,a} \Rightarrow \mathcal{N}(0, C_a)$. Hence, by Slutsky's theorem,

$$\begin{aligned}
&\sqrt{n_a}\left[\begin{pmatrix}\Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_0})\\ \Psi_{\mathrm{PG}}(\widehat{\mu}_{a,s_1})\end{pmatrix} - \begin{pmatrix}\Psi_{\mathrm{PG}}(\mu_{a,s_0})\\ \Psi_{\mathrm{PG}}(\mu_{a,s_1})\end{pmatrix}\right] \\
&\quad \Rightarrow \mathcal{N}(0, \Omega_a),
\end{aligned}$$

where $\Omega_a := A_a C_a A_a^\top \in \mathbb{R}^{2d \times 2d}$.

Now stack the treated and control groups. Multiplying each group expansion by $\sqrt{n/n_a}$ gives

$$\begin{aligned}
&\sqrt{n}\left[\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) - \Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right] \\
&\quad = \begin{pmatrix}\sqrt{n/n_T}\, A_T \dfrac{1}{\sqrt{n_T}} \sum_{i=1}^{n_T} \xi_{i,T} \\ \sqrt{n/n_C}\, A_C \dfrac{1}{\sqrt{n_C}} \sum_{i=1}^{n_C} \xi_{i,C}\end{pmatrix} + o_p(1).
\end{aligned}$$

By Assumption 4.1, the treated and control score sums are independent; within-group dependence between periods remains in C_a. The two group CLTs therefore hold jointly. Since $n/n_a \to \rho_a^{-1}$ for $a \in \{T, C\}$, Slutsky's theorem gives

$$\sqrt{n}\left[\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) - \Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right] \Rightarrow \mathcal{N}(0, \Xi_{\mathrm{G}}),$$

where

$$\Xi_{\mathrm{G}} := \mathrm{diag}\left(\rho_T^{-1}\Omega_T, \rho_C^{-1}\Omega_C\right) \in \mathbb{R}^{4d \times 4d}.$$

This proves the lemma. □

B.2. **Proof of Theorem 4.1.**

Proof. The proof applies the delta method to the joint limit in Lemma 4.1. It first locates the population and sample means in $\mathcal{U}_{\mathrm{PG}}$, where g_{G} is continuously differentiable, then derives the limit of the local discrepancy, shows that the completion outside $\mathcal{E}_n^{\mathrm{G}}$ does not affect this limit, and finally deduces the limit of the Wald statistic.

By Lemma 4.1, $\sqrt{n}\left[\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) - \Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right] \Rightarrow \mathcal{N}(0, \Xi_{\mathrm{G}})$. The consistency $\widehat{\mathbf{m}}_{\mathrm{G}} \to_p \mathbf{m}_{\mathrm{G}}$ comes from the first step of Appendix B.1, which applies Theorem 2.3 of Bhattacharya and Patrangenaru (2003) to each group and period before any CLT is used.

By Assumption 4.2(2), the four population Fréchet means and μ_{ref} belong to the open strongly convex neighborhood $\mathcal{U}_{\mathrm{PG}}$. Hence $\mathbf{m}_{\mathrm{G}} \in \mathcal{U}_{\mathrm{PG}}^4$, where $\mathcal{U}_{\mathrm{PG}}^4 := \mathcal{U}_{\mathrm{PG}} \times \mathcal{U}_{\mathrm{PG}} \times \mathcal{U}_{\mathrm{PG}} \times \mathcal{U}_{\mathrm{PG}}$. Since $\mathcal{U}_{\mathrm{PG}}^4$ is open and $\widehat{\mathbf{m}}_{\mathrm{G}} \to_p \mathbf{m}_{\mathrm{G}}$, $\mathbb{P}(\mathcal{E}_n^{\mathrm{G}}) \to 1$.

On $\mathcal{U}_{\mathrm{PG}}^4$, define $g_{\mathrm{G}}(x_0, x_1, z_0, z_1) := B_{\mathrm{ref}}^{-1} \Pi_{\eta_{x_0 \to \mu_{\mathrm{ref}}}} \left[\mathrm{Log}_{x_0}(x_1) - \Pi_{\eta_{z_0 \to x_0}} \left\{\mathrm{Log}_{z_0}(z_1)\right\}\right]$. Because $\mathcal{U}_{\mathrm{PG}}$ is strongly convex, the unique minimizing geodesic between any two points in $\mathcal{U}_{\mathrm{PG}}$ depends smoothly on its endpoints. Hence, the logarithmic map $(x, y) \mapsto \mathrm{Log}_x(y)$ is smooth on $\mathcal{U}_{\mathrm{PG}} \times \mathcal{U}_{\mathrm{PG}}$, and the parallel transport map $(x, y, v) \mapsto \Pi_{\eta_{x \to y}} v$ is smooth on the set $\{(x, y, v) : x, y \in \mathcal{U}_{\mathrm{PG}}, v \in T_x\mathcal{M}\}$. Since B_{ref}^{-1} is a fixed linear isometry, it follows that g_{G} is continuously differentiable on $\mathcal{U}_{\mathrm{PG}}^4$.

This smoothness claim is local. On a strongly convex coordinate neighborhood, the relevant pairs of endpoints avoid the cut locus, and the exponential map is locally a diffeomorphism. Hence the minimizing geodesic and the logarithmic vector depend smoothly on their endpoints. The parallel-transport equation is a linear ordinary differential equation along this smoothly varying geodesic, so its solution also depends smoothly on the endpoints and the initial vector.

Let $\widetilde{g}_{\mathrm{G}} := g_{\mathrm{G}} \circ \Phi_{\mathrm{G}}^{-1}$. Since the product chart Φ_{G} is a diffeomorphism onto its image, its inverse is smooth, which ensures that $\widetilde{g}_{\mathrm{G}}$ is continuously differentiable in a neighborhood of $\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})$. By definition, $\delta_0^{\mathrm{G}} = g_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}}) = \widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right\}$. Moreover, on the event $\mathcal{E}_n^{\mathrm{G}}$, $\widehat{\delta}_n^{\mathrm{G}} = g_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}}) = \widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})\right\}$. Let $D_{\mathrm{G}} := D\left[g_{\mathrm{G}} \circ \Phi_{\mathrm{G}}^{-1}\right]\left\{\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right\} = D\widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right\}$. By the multivariate delta method, $\sqrt{n}\left[\widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})\right\} - \widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\mathbf{m}_{\mathrm{G}})\right\}\right] \Rightarrow \mathcal{N}\left(0, D_{\mathrm{G}}\Xi_{\mathrm{G}}D_{\mathrm{G}}^{\top}\right)$.

Since $\mathbb{P}(\mathcal{E}_n^{\mathrm{G}}) \to 1$, the arbitrary completion of $\widehat{\delta}_n^{\mathrm{G}}$ on $(\mathcal{E}_n^{\mathrm{G}})^c$ is asymptotically irrelevant. More precisely, the completed estimator $\widehat{\delta}_n^{\mathrm{G}}$ agrees with $\widetilde{g}_{\mathrm{G}}\left\{\Phi_{\mathrm{G}}(\widehat{\mathbf{m}}_{\mathrm{G}})\right\}$ on $\mathcal{E}_n^{\mathrm{G}}$, and hence the two random vectors differ with probability at most $\mathbb{P}((\mathcal{E}_n^{\mathrm{G}})^c) \to 0$. Therefore they have the same limiting distribution at the $\sqrt{n}$ scale.

Therefore, $\sqrt{n}\left(\widehat{\delta}_n^{\mathrm{G}} - \delta_0^{\mathrm{G}}\right) \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{G}})$, where $\Sigma_{\mathrm{G}} := D_{\mathrm{G}} \Xi_{\mathrm{G}} D_{\mathrm{G}}^{\top}$. This proves the asserted CLT. Under H_0^{GDiD}, $\delta_0^{\mathrm{G}} = 0$, and hence $\sqrt{n}\,\widehat{\delta}_n^{\mathrm{G}} \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{G}})$.

It remains to prove the Wald conclusion. Suppose that Σ_{G} is positive definite and $\widehat{\Sigma}_{\mathrm{G}} \to_p \Sigma_{\mathrm{G}}$. By the continuity of the inverse square-root map, $\widehat{\Sigma}_{\mathrm{G}}^{-1/2} \to_p \Sigma_{\mathrm{G}}^{-1/2}$. Thus, under H_0^{GDiD}, Slutsky's theorem gives $\widehat{\Sigma}_{\mathrm{G}}^{-1/2}\sqrt{n}\,\widehat{\delta}_n^{\mathrm{G}} \Rightarrow \mathcal{N}(0, I_d)$. The continuous mapping theorem then yields $W_n^{\mathrm{G}} = n(\widehat{\delta}_n^{\mathrm{G}})^{\top}\widehat{\Sigma}_{\mathrm{G}}^{-1}\widehat{\delta}_n^{\mathrm{G}} \Rightarrow \chi_d^2$. This completes the proof. □

B.3. **Proof of Theorem 4.2.**

Proof. The proof writes the discrepancy through the two groups' sample mean increments, applies the multivariate central limit theorem within each group, combines the independent groups, and then shows that the plug-in covariance estimator is consistent, which yields the Wald limit.

By Assumption 4.3, for each $a \in \{T, C\}$ and $j \in \{0, 1\}$, $O_{i,a,s_j} \in \mathcal{K}_{\mathrm{PT}} \subset \mathcal{U}_{\mathrm{PT}} \subset \mathcal{U}_{\mathrm{ref}}$ almost surely. Hence the common-chart variables $Y_{i,a,s_j} = \Psi_{\mathrm{ref}}(O_{i,a,s_j}) = B_{\mathrm{ref}}^{-1}\,\mathrm{Log}_{\mu_{\mathrm{ref}}}(O_{i,a,s_j})$ are well-defined.

Since $\mathcal{K}_{\mathrm{PT}}$ is compact and Ψ_{ref} is continuous on $\mathcal{U}_{\mathrm{PT}}$, the chart image $\Psi_{\mathrm{ref}}(\mathcal{K}_{\mathrm{PT}})$ is compact in $\mathbb{R}^d$. Therefore, the random vectors Y_{i,a,s_j} are bounded. Consequently, $\Delta_{i,a} = Y_{i,a,s_1} - Y_{i,a,s_0}$ is bounded, and in particular $\mathbb{E}\|\Delta_a\|^2 < \infty$.

By Assumption 4.1, for each group $a \in \{T, C\}$, the paired trajectories $\mathbf{O}_{i,a} = (O_{i,a,s_0}, O_{i,a,s_1})$ are i.i.d. Hence $\Delta_{i,a}$ is an i.i.d. sequence within group a. Let $m_a := \mathbb{E}[\Delta_a]$ and $V_a := \mathrm{Var}(\Delta_a)$ for $a \in \{T, C\}$. By the multivariate central limit theorem, $\sqrt{n_a}\left(\overline{\Delta}_a - m_a\right) \Rightarrow \mathcal{N}(0, V_a)$.

By definition, $\delta_0^{\mathrm{T}} = m_T - m_C$ and $\widehat{\delta}_n^{\mathrm{T}} - \delta_0^{\mathrm{T}} = (\overline{\Delta}_T - m_T) - (\overline{\Delta}_C - m_C)$. Therefore, $\sqrt{n}\left(\widehat{\delta}_n^{\mathrm{T}} - \delta_0^{\mathrm{T}}\right) = \sqrt{\frac{n}{n_T}}\sqrt{n_T}(\overline{\Delta}_T - m_T) - \sqrt{\frac{n}{n_C}}\sqrt{n_C}(\overline{\Delta}_C - m_C)$. The treated and control samples are independent by Assumption 4.1, and $\frac{n_a}{n} \to \rho_a \in (0, 1)$ for $a \in \{T, C\}$. Hence, by Slutsky's theorem, $\sqrt{n}\left(\widehat{\delta}_n^{\mathrm{T}} - \delta_0^{\mathrm{T}}\right) \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{T}})$, where $\Sigma_{\mathrm{T}} = \rho_T^{-1} V_T + \rho_C^{-1} V_C$. Under H_0^{TDiD}, $\delta_0^{\mathrm{T}} = 0$, so $\sqrt{n}\,\widehat{\delta}_n^{\mathrm{T}} \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{T}})$.

We next show consistency of the plug-in covariance estimator. Since $\mathbb{E}\|\Delta_a\|^2 < \infty$, the weak law of large numbers gives $\frac{1}{n_a}\sum_{i=1}^{n_a} \Delta_{i,a}\Delta_{i,a}^{\top} \to_p \mathbb{E}[\Delta_a \Delta_a^{\top}]$, and $\overline{\Delta}_a \to_p m_a$. Thus $\widehat{V}_a = \frac{1}{n_a}\sum_{i=1}^{n_a}(\Delta_{i,a} - \overline{\Delta}_a)(\Delta_{i,a} - \overline{\Delta}_a)^{\top} \to_p V_a$. Since $n/n_a \to \rho_a^{-1}$, $\widehat{\Sigma}_{\mathrm{T}} = \frac{n}{n_T}\widehat{V}_T + \frac{n}{n_C}\widehat{V}_C \to_p \rho_T^{-1} V_T + \rho_C^{-1} V_C = \Sigma_{\mathrm{T}}$.

Finally, suppose that Σ_{T} is positive definite. Under H_0^{TDiD}, $\sqrt{n}\,\widehat{\delta}_n^{\mathrm{T}} \Rightarrow \mathcal{N}(0, \Sigma_{\mathrm{T}})$, and we have just shown that $\widehat{\Sigma}_{\mathrm{T}} \to_p \Sigma_{\mathrm{T}}$. By the continuity of the inverse square-root map on the cone of positive definite matrices, $\widehat{\Sigma}_{\mathrm{T}}^{-1/2} \to_p \Sigma_{\mathrm{T}}^{-1/2}$. Slutsky's theorem implies $\widehat{\Sigma}_{\mathrm{T}}^{-1/2}\sqrt{n}\,\widehat{\delta}_n^{\mathrm{T}} \Rightarrow \mathcal{N}(0, I_d)$. Therefore, by the continuous mapping theorem, $W_n^{\mathrm{T}} = n(\widehat{\delta}_n^{\mathrm{T}})^{\top}\widehat{\Sigma}_{\mathrm{T}}^{-1}\widehat{\delta}_n^{\mathrm{T}} \Rightarrow \chi_d^2$. This completes the proof. □

Appendix C. Proof of MCiC identification and estimation

C.1. **Proof of Proposition 5.1.**

Proof. The proof first shows that $T^{(0)}_{0\to1}$ is well-defined and measurable and writes the four chart laws as pushforwards of Q_C and Q_T. Statements (1)–(5) then follow in order from these representations and Assumption 5.1. By Assumption 5.1(2), $h_0 : \mathcal{Z} \to \mathcal{Y}_0$ is bijective and $h_0^{-1} : \mathcal{Y}_0 \to \mathcal{Z}$ is measurable. Since $h_1 = \Psi_{\mathrm{ref}} \circ \widetilde{h}_1$ is measurable and $h_1(\mathcal{Z}) \subset \mathcal{Y}_1$, the map $T^{(0)}_{0\to1} := h_1 \circ h_0^{-1}$ is a well-defined measurable map from $\mathcal{Y}_0$ to $\mathcal{Y}_1$. By Assumption 5.1(1) and the definitions of h_0 and h_1, we have $Y_{C,0} = h_0(Z_C)$, $Y_{C,1} = h_1(Z_C)$, $Y_{T,0} = h_0(Z_T)$, and $Y_{T,1}(0) = h_1(Z_T)$ almost surely. Hence $\nu_{C,0} = h_{0\#}Q_C$, $\nu_{C,1} = h_{1\#}Q_C$, $\nu_{T,0} = h_{0\#}Q_T$, and $\nu^{(0)}_{T,1} = h_{1\#}Q_T$.

Because $T^{(0)}_{0\to1} \circ h_0 = h_1$ on $\mathcal{Z}$, we obtain $(T^{(0)}_{0\to1})_\#\nu_{C,0} = (T^{(0)}_{0\to1})_\# h_{0\#}Q_C = (T^{(0)}_{0\to1} \circ h_0)_\# Q_C = h_{1\#}Q_C = \nu_{C,1}$, which proves (1). For (2), Assumption 5.1(4) gives $h_{0\#}Q_C \ll \lambda^d$. Since $\nu_{C,0} = h_{0\#}Q_C$, it follows immediately that $\nu_{C,0} \ll \lambda^d$, which proves (2). Take any positive integer r and any cycle $y_1, \dots, y_r, y_{r+1} \equiv y_1 \in \mathcal{Y}_0$. By Assumption 5.1(2), $z_j := h_0^{-1}(y_j) \in \mathcal{Z}$ is well-defined for each j. Since $T^{(0)}_{0\to1}(y_j) = h_1(z_j)$, Assumption 5.1(3) yields $\sum_{j=1}^r \left\langle y_j, T^{(0)}_{0\to1}(y_j) - T^{(0)}_{0\to1}(y_{j+1}) \right\rangle = \sum_{j=1}^r \langle h_0(z_j), h_1(z_j) - h_1(z_{j+1}) \rangle \geq 0$. Thus $T^{(0)}_{0\to1}$ is cyclically monotone, proving (3).

Next, since $Q_T \ll Q_C$ by Assumption 5.1(5), pushforward by the common map h_0 gives $\nu_{T,0} \ll \nu_{C,0}$. Moreover, since $h_0(\mathcal{Z}) = \mathcal{Y}_0$ and $\mathcal{Y}_0$ is compact, and so closed, both $\nu_{C,0}$ and $\nu_{T,0}$ are supported on $\mathcal{Y}_0$. Thus $\mathcal{S}_{C,0} \subset \mathcal{Y}_0$ and $\mathcal{S}_{T,0} \subset \mathcal{Y}_0$. Together with $\nu_{T,0} \ll \nu_{C,0}$, this implies $\mathcal{S}_{T,0} \subset \mathcal{S}_{C,0} \subset \mathcal{Y}_0$, which proves (4). Finally, the same argument used for (1) with Q_T in place of Q_C gives $(T^{(0)}_{0\to1})_\#\nu_{T,0} = (T^{(0)}_{0\to1})_\# h_{0\#}Q_T = (T^{(0)}_{0\to1} \circ h_0)_\# Q_T = h_{1\#}Q_T = \nu^{(0)}_{T,1}$, which proves (5). □

C.2. **Proof of Theorem 5.1.**

Proof. The proof identifies the three chart laws, applies Brenier's theorem to the two control laws, uses Proposition 5.1 to show that the resulting Brenier map agrees with $T^{(0)}_{0\to1}$ almost everywhere under the control and treated baseline laws, and maps the counterfactual chart law back to $\mathcal{M}$. By the observation rule, $O_{C,0} = O_{C,0}(0)$, $O_{C,1} = O_{C,1}(0)$, and $O_{T,0} = O_{T,0}(0)$. Therefore, the manifold-valued laws $P_{C,0}$, $P_{C,1}$, and $P_{T,0}$ are identified from the observed distribution. Since Ψ_{ref} is known, the tangent-space laws $\nu_{C,0} = (\Psi_{\mathrm{ref}})_\# P_{C,0}$, $\nu_{C,1} = (\Psi_{\mathrm{ref}})_\# P_{C,1}$, and $\nu_{T,0} = (\Psi_{\mathrm{ref}})_\# P_{T,0}$ are also identified.

By Assumption 5.1, the set $\mathcal{K}_{\mathrm{MCiC}}$ is compact, and therefore its chart image $\mathcal{Y}_{\mathrm{ref}} = \Psi_{\mathrm{ref}}(\mathcal{K}_{\mathrm{MCiC}})$ is compact in $\mathbb{R}^d$. Since $O_{C,0}$ and $O_{C,1}$ take values in $\mathcal{K}_{\mathrm{MCiC}}$ almost surely under Assumption

5.1, the laws $\nu_{C,0}$ and $\nu_{C,1}$ are supported on the compact set $\mathcal{Y}_{\mathrm{ref}}$. Hence, $\nu_{C,0}, \nu_{C,1} \in \mathcal{P}_2(\mathbb{R}^d)$. Moreover, Proposition 5.1(2) gives $\nu_{C,0} \ll \lambda^d$. Therefore, Brenier's theorem (Brenier (1991), Villani (2003)) applies to the pair $(\nu_{C,0}, \nu_{C,1})$: for the squared Euclidean cost, there exists a $\nu_{C,0}$-almost everywhere unique optimal transport map $B^C_{0\to1} : \mathbb{R}^d \to \mathbb{R}^d$ pushing $\nu_{C,0}$ forward to $\nu_{C,1}$. This map is determined by the observed laws $\nu_{C,0}$ and $\nu_{C,1}$, and is therefore identified.

It remains to relate this Brenier map to the common untreated transition map $T^{(0)}_{0\to1}$. By Proposition 5.1(1), $\nu_{C,1} = (T^{(0)}_{0\to1})_{\#}\nu_{C,0}$, so $T^{(0)}_{0\to1}$ transports $\nu_{C,0}$ to $\nu_{C,1}$. By Proposition 5.1(3), $T^{(0)}_{0\to1}$ is cyclically monotone. Hence the coupling $(\mathrm{id}, T^{(0)}_{0\to1})_{\#}\nu_{C,0}$ is optimal for the squared Euclidean cost. Since the optimal transport map from $\nu_{C,0}$ to $\nu_{C,1}$ is $\nu_{C,0}$-almost everywhere unique, we obtain $B^C_{0\to1} = T^{(0)}_{0\to1}$ $\nu_{C,0}$-almost everywhere. By Proposition 5.1(4), $\nu_{T,0} \ll \nu_{C,0}$. Therefore, the equality $B^C_{0\to1} = T^{(0)}_{0\to1}$ also holds $\nu_{T,0}$-almost everywhere. Combining this with Proposition 5.1(5), we obtain $(B^C_{0\to1})_{\#}\nu_{T,0} = (T^{(0)}_{0\to1})_{\#}\nu_{T,0} = \nu^{(0)}_{T,1}$. Thus the untreated counterfactual distribution on the common tangent-space chart is identified. Finally, since $\nu^{(0)}_{T,1}$ is supported on $\mathcal{Y}_{\mathrm{ref}}$ and Ψ^{-1}_{ref} is well-defined on $\mathcal{Y}_{\mathrm{ref}}$, the corresponding counterfactual distribution on the manifold is $P^{(0)}_{T,1} = (\Psi^{-1}_{\mathrm{ref}})_{\#}\nu^{(0)}_{T,1} = (\Psi^{-1}_{\mathrm{ref}})_{\#}\big((B^C_{0\to1})_{\#}\nu_{T,0}\big) = (\Psi^{-1}_{\mathrm{ref}})_{\#}\big((T^{(0)}_{0\to1})_{\#}\nu_{T,0}\big)$. Hence $P^{(0)}_{T,1}$ is identified. □

C.3. **Proof of Lemma 5.1.**

Proof. By Assumption 2.1 and the Hopf–Rinow theorem, $\mathrm{Exp}_{\mu_{\mathrm{ref}}}$ is defined and smooth on the whole tangent space $T_{\mu_{\mathrm{ref}}}\mathcal{M}$. Hence $u \mapsto \mathrm{Exp}_{\mu_{\mathrm{ref}}}(B_{\mathrm{ref}}u)$ is smooth on $\mathbb{R}^d$, and C_Ψ is finite because its derivative is continuous and $\mathcal{Y}$ is compact. For $u, v \in \mathcal{Y}$, the curve $c(t) := \mathrm{Exp}_{\mu_{\mathrm{ref}}}[B_{\mathrm{ref}}\{(1-t)u+tv\}]$, $t \in [0,1]$, joins the two image points, and the segment $(1-t)u + tv$ stays in $\mathcal{Y}$ because $\mathcal{Y}$ is convex. Therefore

$$d_{\mathcal{M}}\{c(0), c(1)\} \le \mathrm{length}(c) = \int_0^1 \left\| D\,\mathrm{Exp}_{\mu_{\mathrm{ref}}}|_{B_{\mathrm{ref}}\{(1-t)u+tv\}} B_{\mathrm{ref}}(v-u) \right\| \mathrm{d}t \le C_\Psi \|u - v\|. \qquad \square$$

C.4. **Proof of Theorem 5.2.**

Proof. We prove the following bound, with the sample sizes of Theorem 5.2 and the constants K_1 and K_2 obtained in Step 2:

$$\mathbb{E}W_{2,\mathcal{M}}\Big(\widehat{P}^{(0),1\mathrm{NN}}_{T,1,m}, P^{(0)}_{T,1}\Big) \le C_\Psi \Bigg[K_1 \log(en) \begin{cases} (n\wedge\ell)^{-1/2}, & d=1,\\ (n\wedge\ell)^{-1/2}\log\{e(n\wedge\ell)\}, & d=2,\\ (n\wedge\ell)^{-1/d}, & d\ge 3, \end{cases} + K_2 L \begin{cases} m^{-1/2}, & d=1,\\ m^{-1/2}\log(em), & d=2,\\ m^{-1/d}, & d\ge 3. \end{cases} \Bigg].$$

The proof splits the chart Wasserstein distance into a 1NN map-error term (I) and a treated empirical-distribution term (II) in Step 1. Step 2 collects the two results of Manole et al. (2024) that the proof uses. Step 3 bounds (I) through the 1NN map-risk bound, Step 4 bounds (II) through the Lipschitz property of the true map and an empirical Wasserstein bound, and Step 5 transfers the combined bound from the chart to $\mathcal{M}$ through the inverse chart.

All measures and maps in this proof are expressed in the common normalized chart coordinates of Assumption 5.2, in which the supports of $\nu_{C,0}$, $\nu_{C,1}$, and $\nu_{T,0}$ are the unit cube $[0,1]^d$. Throughout the proof, the cube refers to $[0,1]^d$. Independently, $Y_{C,0,1},\dots,Y_{C,0,n}$ are i.i.d. from $\nu_{C,0}$, $Y_{C,1,1},\dots,Y_{C,1,\ell}$ are i.i.d. from $\nu_{C,1}$, and $Y_{T,0,1},\dots,Y_{T,0,m}$ are i.i.d. from $\nu_{T,0}$. Each chart density lies between c and C on $[0,1]^d$, the potential of the Brenier map $B^C_{0\to1}=\nabla\phi$ from $\nu_{C,0}$ to $\nu_{C,1}$ satisfies $L^{-1}I_d \preceq \nabla^2\phi \preceq LI_d$, and, by Lemma 5.1, the normalized inverse chart is C_Ψ-Lipschitz on the cube. Write $\widehat{\nu}_{T,0,m}=m^{-1}\sum_{k=1}^m \delta_{Y_{T,0,k}}$. For the selected empirical control OT mass matrix $(\widehat{\pi}_{ij})$, its barycentric image at source point $Y_{C,0,i}$ is

$$\widehat{B}^C_{0\to1,n,\ell}(Y_{C,0,i}) := \frac{\sum_{j=1}^\ell \widehat{\pi}_{ij} Y_{C,1,j}}{\sum_{j=1}^\ell \widehat{\pi}_{ij}}.$$

Step 1. Decomposition into (I) and (II). Set $i_n(y) = \min\operatorname{argmin}_{1\le i\le n}\|y - Y_{C,0,i}\|$, $\pi_n(y) = Y_{C,0,i_n(y)}$, and $\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell}(y) = \widehat{B}^C_{0\to1,n,\ell}\{\pi_n(y)\}$. Theorem 5.1 identifies $P^{(0)}_{T,1} = (\Psi_{\mathrm{ref}}^{-1})_\#(B^C_{0\to1})_\#\nu_{T,0}$, while the estimator is $\widehat{P}^{(0),1\mathrm{NN}}_{T,1,m} = (\Psi_{\mathrm{ref}}^{-1})_\#(\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell})_\#\widehat{\nu}_{T,0,m}$. By the triangle inequality for W_2, the

chart error between these two laws splits into a 1NN map-error term (I) and a treated empirical-distribution term (II),

$$W_2\Big\{(\widehat{B}^{C,\mathrm{1NN}}_{0\to1,n,\ell})_{\#}\widehat{\nu}_{T,0,m},(B^C_{0\to1})_{\#}\nu_{T,0}\Big\} \le \underbrace{W_2\Big\{(\widehat{B}^{C,\mathrm{1NN}}_{0\to1,n,\ell})_{\#}\widehat{\nu}_{T,0,m},(B^C_{0\to1})_{\#}\widehat{\nu}_{T,0,m}\Big\}}_{\text{(I)}} + \underbrace{W_2\Big\{(B^C_{0\to1})_{\#}\widehat{\nu}_{T,0,m},(B^C_{0\to1})_{\#}\nu_{T,0}\Big\}}_{\text{(II)}}.$$

Term (I) compares the estimated and true maps on the same treated sample. Term (II) is the error of the treated empirical law after it and the population law are pushed through the true map.

Step 2. Two results of Manole et al. (2024). The cube is convex, has nonempty interior, and satisfies the standard-set condition in Manole et al. (2024). Together with the density and Hessian bounds of Assumption 5.2, this allows us to apply Proposition 15 of Manole et al. (2024) to the empirical plan's barycentric projection with its 1NN extension. The constant in that proposition depends only on d, L, c, and C, because the standard-set constants of the cube depend only on d. We write it as $(c/C)K_1^2$, so that K_1 also absorbs the density-ratio bound used in Step 3, and obtain

$$\mathbb{E}\int \|\widehat{B}^{C,\mathrm{1NN}}_{0\to1,n,\ell}(y)-B^C_{0\to1}(y)\|^2\,\nu_{C,0}(\mathrm{d}y) \le (c/C)K_1^2\log^2(en)\begin{cases}(n\wedge\ell)^{-1}, & d=1,\\ (n\wedge\ell)^{-1}\log^2\{e(n\wedge\ell)\}, & d=2,\\ (n\wedge\ell)^{-2/d}, & d\ge3.\end{cases}$$

The leading logarithm is in the source sample size n, as stated in Proposition 15 of Manole et al. (2024).

For the treated empirical law, Corollary 8 of Manole et al. (2024) bounds the expected squared 2-Wasserstein distance between an empirical law on the cube and its population law. We apply it with both population laws equal to $\nu_{T,0}$, so that its optimal map is the identity, whose potential is $y\mapsto\|y\|^2/2$ and whose Hessian is I_d, and the treated density is bounded above and below on the cube. The corollary gives a constant K_2^2, depending only on d, c, and C, for which

$$\mathbb{E}W_2^2(\widehat{\nu}_{T,0,m},\nu_{T,0}) \le K_2^2\begin{cases}m^{-1}, & d=1,\\ m^{-1}\log^2(em), & d=2,\\ m^{-2/d}, & d\ge3.\end{cases}$$

Writing the logarithms as $\log(en)$, $\log\{e(n \wedge \ell)\}$, and $\log(em)$ also covers small samples, because each risk on the left is at most d, the squared diameter of the cube, and this trivial bound is absorbed into the constants.

Step 3. Bound on (I). For term (I), let $f_{T|C} := \mathrm{d}\nu_{T,0}/\mathrm{d}\nu_{C,0}$ denote the density ratio. Since both densities lie between c and C on the cube, $\|f_{T|C}\|_\infty \le C/c$. The measure $m^{-1}\sum_{k=1}^m \delta_{(\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell}(Y_{T,0,k}),B^C_{0\to1}(Y_{T,0,k}))}$ is a coupling of its two empirical image laws. Thus, conditional on the two control samples, $W_2^2\{(\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell})_\#\widehat{\nu}_{T,0,m}, (B^C_{0\to1})_\#\widehat{\nu}_{T,0,m}\} \le m^{-1}\sum_k \|\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell}(Y_{T,0,k}) - B^C_{0\to1}(Y_{T,0,k})\|^2$. The treated sample is independent of the two control samples, so taking expectations, applying Jensen's inequality, and using $\nu_{T,0} = f_{T|C}\nu_{C,0}$ yields

$$\begin{aligned}\mathbb{E}W_2\{(\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell})_\#\widehat{\nu}_{T,0,m}, (B^C_{0\to1})_\#\widehat{\nu}_{T,0,m}\} &\le \left\{\mathbb{E}\int \|\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell} - B^C_{0\to1}\|^2 \,\mathrm{d}\nu_{T,0}\right\}^{1/2} \\ &\le \|f_{T|C}\|_\infty^{1/2}\left\{\mathbb{E}\int \|\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell} - B^C_{0\to1}\|^2 \,\mathrm{d}\nu_{C,0}\right\}^{1/2}.\end{aligned}$$

Since $\|f_{T|C}\|_\infty^{1/2}(c/C)^{1/2} \le 1$, taking square roots in the map-risk bound of Step 2 bounds the expectation of term (I) by

$$K_1 \log(en)\begin{cases}(n \wedge \ell)^{-1/2}, & d = 1,\\ (n \wedge \ell)^{-1/2}\log\{e(n \wedge \ell)\}, & d = 2,\\ (n \wedge \ell)^{-1/d}, & d \ge 3.\end{cases}$$

The coupling uses the same treated observations on both sides and never requires $\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell}$ to be Lipschitz. Its action on those observations is also identical to the empirical-support projection implementation: source points are almost surely distinct under the positive Lebesgue density, and the same smallest-index 1NN rule is used for π_n and $\widehat{B}^{C,1\mathrm{NN}}_{0\to1,n,\ell}$.

Step 4. Bound on (II). Term (II) uses the Lipschitz continuity of the true map. The true map has a smooth version on the whole cube because $B^C_{0\to1} = \nabla\phi$ there. Its upper Hessian bound and convexity of the cube imply, for u, v in the cube,

$$\|B^C_{0\to1}(u) - B^C_{0\to1}(v)\| = \left\|\int_0^1 \nabla^2\phi\{u + t(v-u)\}(v-u)\,\mathrm{d}t\right\| \le L\|u - v\|.$$

Push a coupling of $\widehat{\nu}_{T,0,m}$ and $\nu_{T,0}$ through $(B^C_{0\to1}, B^C_{0\to1})$. By this L-Lipschitz property, $W_2\{(B^C_{0\to1})_\#\widehat{\nu}_{T,0,m}, (B^C_{0\to1})_\#\nu_{T,0}\} \le LW_2(\widehat{\nu}_{T,0,m}, \nu_{T,0})$. Jensen's inequality and the empirical W_2^2

bound of Step 2 then bound the expectation of term (II) by

$$K_2 L \begin{cases} m^{-1/2}, & d = 1, \\ m^{-1/2} \log(em), & d = 2, \\ m^{-1/d}, & d \geq 3. \end{cases}$$

Step 5. Transfer to $\mathcal{M}$. Both image laws are supported on the cube. Each value of the 1NN map is a barycentric image, which is an average of control post-period observations in the cube. For the population law, $(B^C_{0\to 1})_{\#}\nu_{C,0} = \nu_{C,1}$ is supported on the cube and $\nu_{T,0} \ll \nu_{C,0}$, so $(B^C_{0\to 1})_{\#}\nu_{T,0}$ is also supported on the cube. The normalized inverse chart is C_Ψ-Lipschitz there. Apply that inverse chart to a Euclidean coupling between $(\widehat{B}^{C,\mathrm{1NN}}_{0\to 1,n,\ell})_{\#}\widehat{\nu}_{T,0,m}$ and $(B^C_{0\to 1})_{\#}\nu_{T,0}$ and take an infimum. This gives

$$W_{2,\mathcal{M}}\Big(\widehat{P}^{(0),\mathrm{1NN}}_{T,1,m}, P^{(0)}_{T,1}\Big) \leq C_\Psi W_2\Big\{(\widehat{B}^{C,\mathrm{1NN}}_{0\to 1,n,\ell})_{\#}\widehat{\nu}_{T,0,m}, (B^C_{0\to 1})_{\#}\nu_{T,0}\Big\}.$$

Taking expectations in this reduction, bounding the right-hand side by C_Ψ times the sum of (I) and (II) from Step 1, and applying the bounds of Steps 3 and 4 gives

$$\mathbb{E}W_{2,\mathcal{M}}\Big(\widehat{P}^{(0),\mathrm{1NN}}_{T,1,m}, P^{(0)}_{T,1}\Big) \leq C_\Psi \Bigg[K_1 \log(en) \begin{cases} (n\wedge\ell)^{-1/2}, & d = 1, \\ (n\wedge\ell)^{-1/2} \log\{e(n\wedge\ell)\}, & d = 2, \\ (n\wedge\ell)^{-1/d}, & d \geq 3, \end{cases}$$
$$+ K_2 L \begin{cases} m^{-1/2}, & d = 1, \\ m^{-1/2} \log(em), & d = 2, \\ m^{-1/d}, & d \geq 3. \end{cases} \Bigg].$$

When $n \asymp \ell \asymp m \asymp N$, the first term has orders $N^{-1/2}\log N$, $N^{-1/2}(\log N)^2$, and $N^{-1/d}\log N$ for $d = 1$, $d = 2$, and $d \geq 3$, respectively. The second term has orders $N^{-1/2}$, $N^{-1/2}\log N$, and $N^{-1/d}$ in those dimensions. Thus the sum has the three rates stated in Theorem 5.2. □

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